



TNB Low Voltage (TNB LV) Transfer of Technology & Knowledge (ToTK)

22nd July 2024 - 25th July 2024

**Your Technology Partner in Geospatial,
Data and Information Technology**

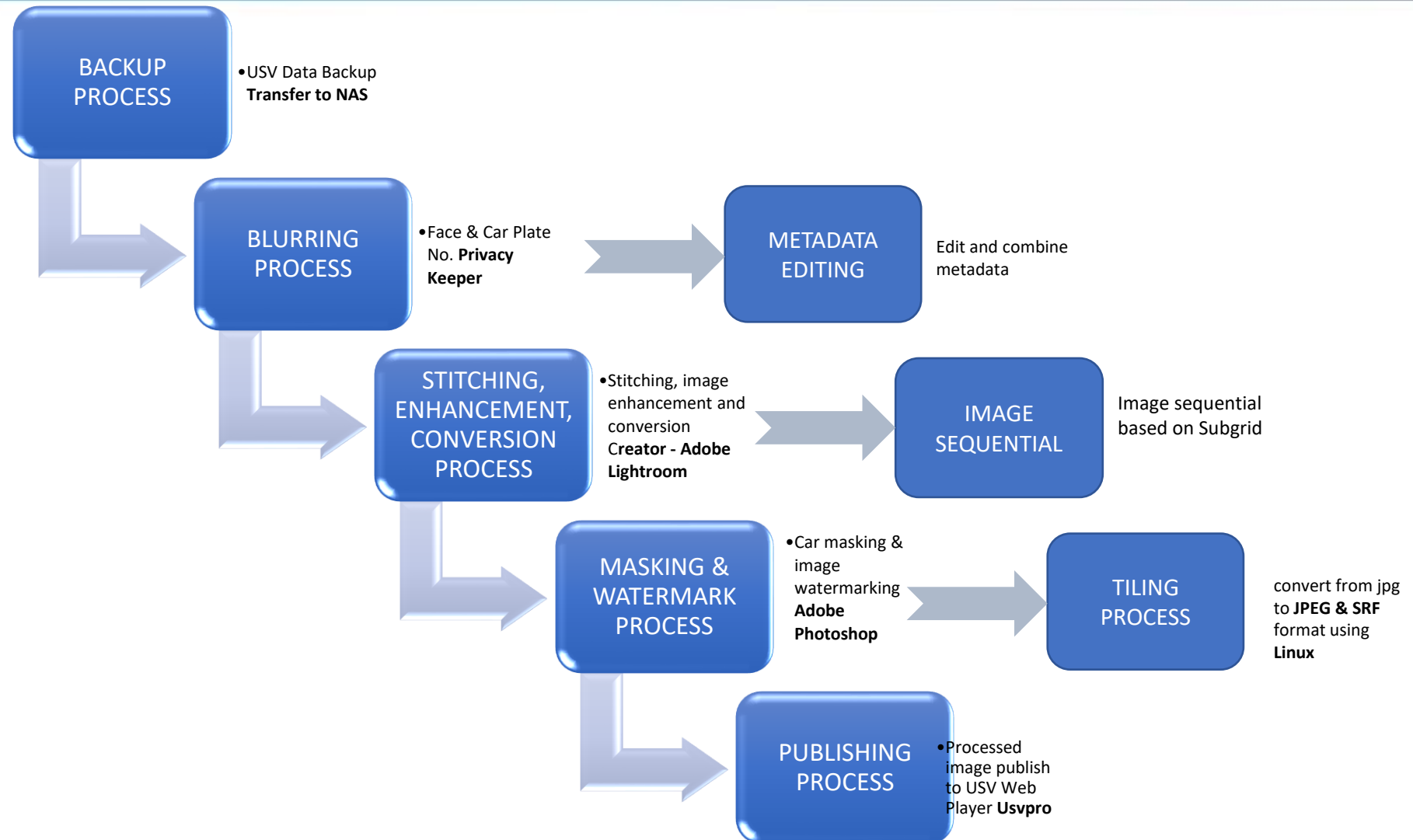
TNB 1577/2021 TNB - GIS DATA PRODUCTION OF LV ASSETS AND NETWORK INCLUDING CUSTOMER METER USING MOBILE MAPPING, DISTRIBUTION NETWORK – MAINHEAD K (TANGKAK & SEGAMAT) & MAINHEAD N (PASIR GUDANG)

TRANSFER OF TECHNOLOGY AND KNOWLEDGE
(TOTK) PROGRAM

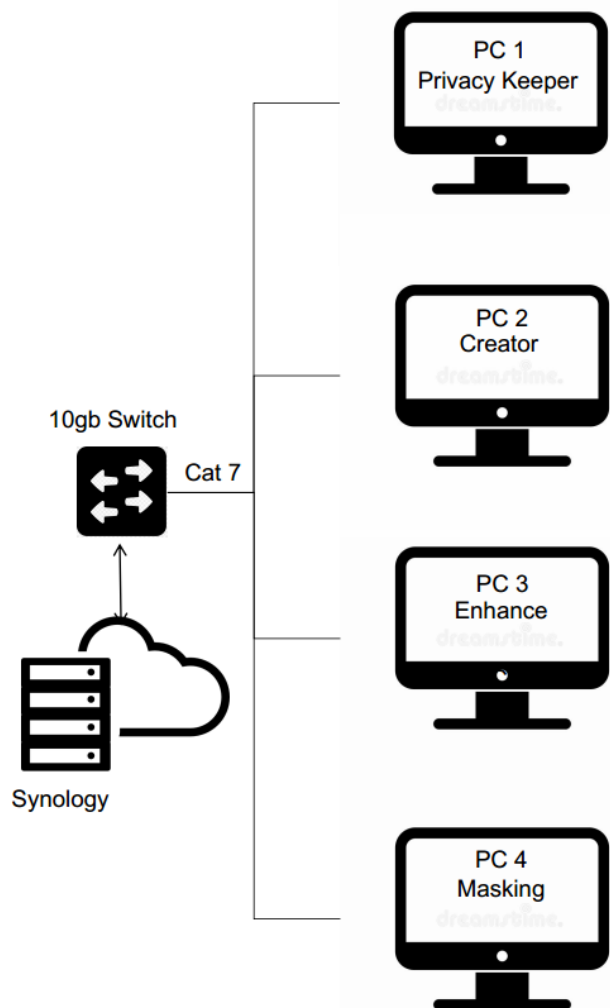
PHASE II :DATA PROCESSING

PREPARED BY : MOHAMAD FARIZ FARHAN BIN
MOHAMAD YAZID

DATA PROCESSING PLAN



Processing Room Diagram



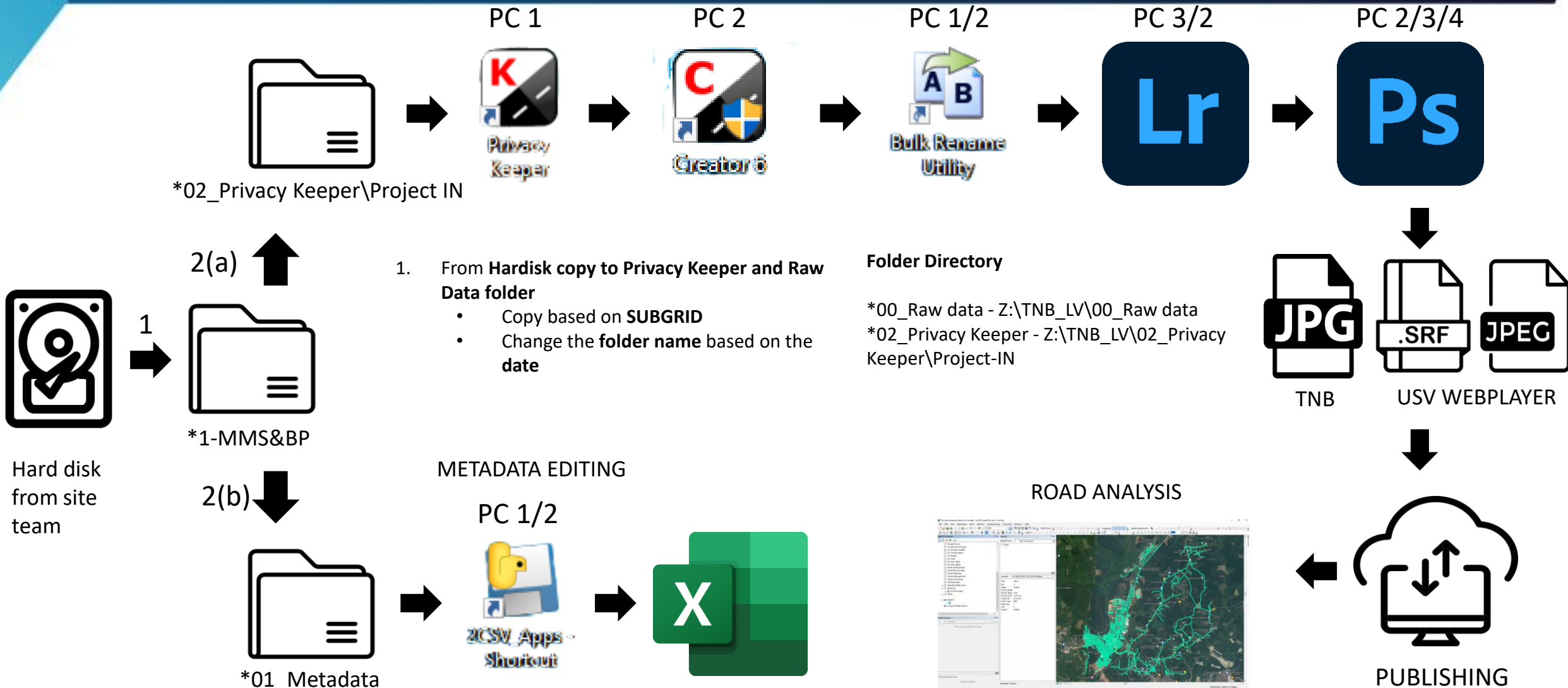
Windows version: Microsoft Windows 10 Pro 64-bit
Processor: 11th Gen Intel(R) Core(TM) i9-11900KF @ 3.50GHz
Memory Total: 63.85 GB Available: 50.48 GB
Graphics Card: NVIDIA GeForce RTX 3050

Windows version: Microsoft Windows 10 Pro 64-bit
Processor: 11th Gen Intel(R) Core(TM) i9-11900KF @ 3.50GHz
Memory Total: 63.85 GB Available: 50.48 GB
Graphics Card: NVIDIA GeForce RTX 2060

Windows version: Microsoft Windows 10 Pro 64-bit
Processor: 11th Gen Intel(R) Core(TM) i9-11900KF @ 3.50GHz
Memory Total: 63.85 GB Available: 50.48 GB
Graphics Card: NVIDIA GeForce RTX 3050

Windows version: Microsoft Windows 10 Pro 64-bit
Processor: 11th Gen Intel(R) Core(TM) i9-11900KF @ 3.50GHz
Memory Total: 63.85 GB Available: 50.48 GB
Graphics Card: NVIDIA GeForce RTX 3050

DATA PROCESSING PLAN



DATA TRANSFER & SORTING

1. From hard disk to NAS>Z:/Location/example/00_raw_data

- Copy all new data based on **GRID FOLDER**
- **Raw data & Privacy keeper** folder

2 types of data :
a.MMS
b.Backpack

a

Desktop > PROCESSING TOTK > Data Processing > 00_Raw data		
Name	Date modified	
N93E70_BP_20220630	30/4/2024 11:40 AM	
N93E70_MMS_20220904	30/4/2024 11:40 AM	

b

Desktop > PROCESSING TOTK > Data Processing > 02_Privacy Keeper >		
Name	Date modified	
20220904	7/9/2022 3:58 PM	
BP_20220630	4/7/2022 12:38 PM	

Time and date tour

NG TOTK > Data Processing > 02_Privacy Keeper > 20220904		
Name	Date modified	Type
003485-20220904-144310	7/9/2022 3:59 PM	File folder
003485-20220904-144428	7/9/2022 3:58 PM	File folder
003485-20220904-144543	7/9/2022 3:58 PM	File folder
003485-20220904-144647	7/9/2022 3:58 PM	File folder
003485-20220904-145542	7/9/2022 3:58 PM	File folder

Total data of each camera including metadata

02_Privacy Keeper > 20220904 > 003485-20220904-144310 >		
Name	Date modified	Type
1	7/9/2022 3:59 PM	File folder
2	7/9/2022 3:59 PM	File folder
3	7/9/2022 3:58 PM	File folder
4	7/9/2022 3:58 PM	File folder
5	7/9/2022 3:58 PM	File folder
6	7/9/2022 3:58 PM	File folder
7	7/9/2022 3:58 PM	File folder

2. Validate and verify data of MMS & backpack

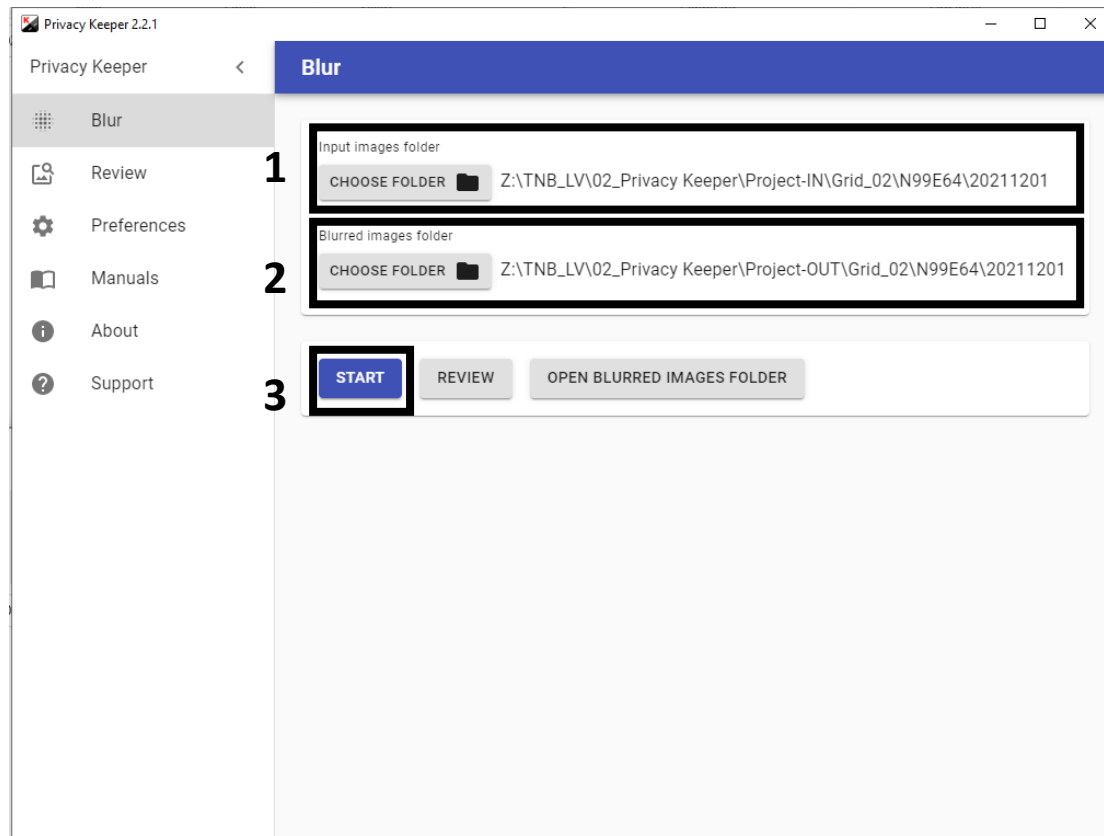
- All data on the **same day tour** must be match
- include **6 folder** camera and **1 folder** of metadata

BLURING PROCESS USING PRIVACY KEEPER

THE PROCESS OF BLURING



PRIVACY KEEPER



Open Privacy Keeper

1. For Input - Click Choose Folder> **Z:\TNB_LV\02_Privacy Keeper\Project-IN\GRID\SUBGRID\DATE**
2. For Output - Click Choose Folder> **Z:\TNB_LV\02_Privacy Keeper\Project-OUT\GRID\SUBGRID\DATE**
3. Click START then wait for the process finished

Output Example



STITCHING PROCESS USING CREATOR 6

THE PROCESS OF PANORAMIC IMAGES



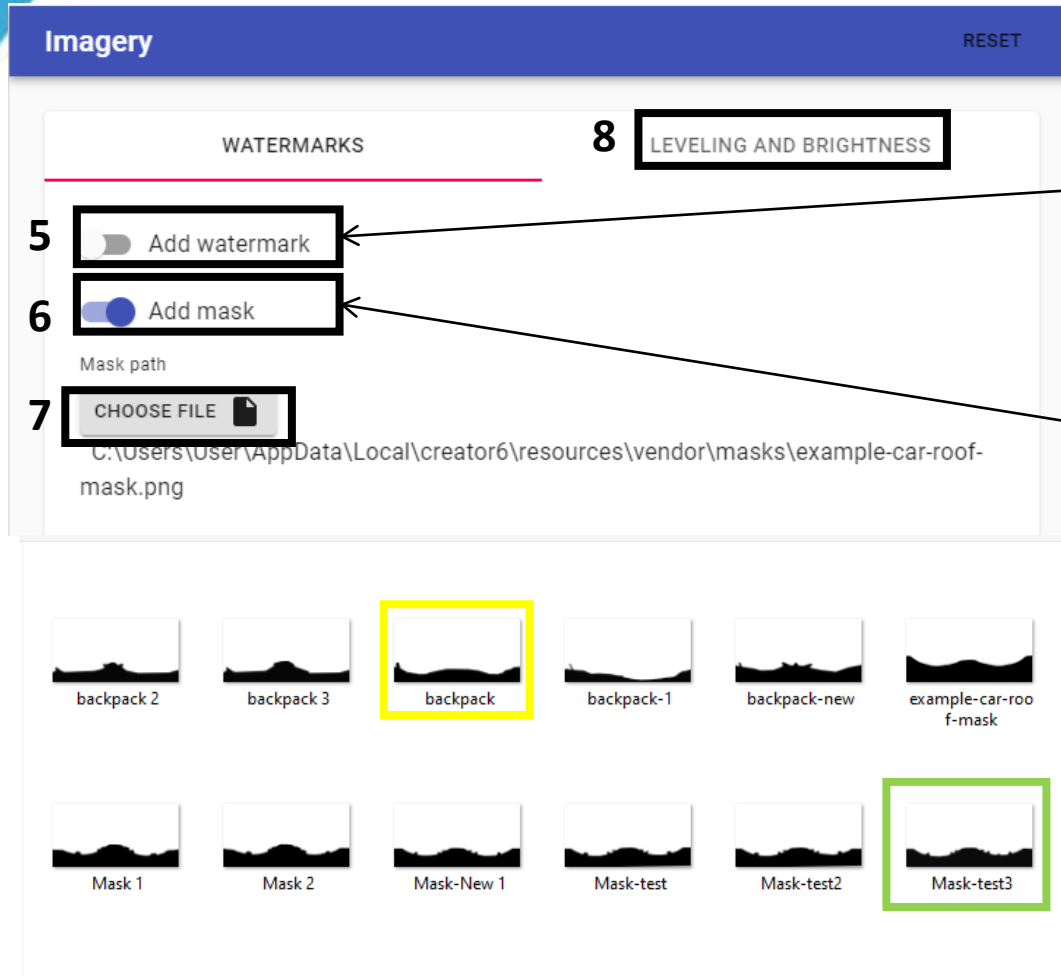
Projects			
			MORE : 1 ADD
Project	Streetviews	Created date ↓	
N99E64-20211201	584	2021-12-03 09:01:14	⋮

Projects - Add	
CANCEL 4 SAVE	
2	Project folder CHOOSE FOLDER Y:\TNB_LV\02_Privacy Keeper\Project-OUT\Grid_02\N99E65\BP_20211123
3	Project name N99E65_BP_20211123

On Creator 6

1. Click **ADD** button
2. Choose Project folder -make sure at **project out PK** (Example Y:\TNB_LV\02_Privacy Keeper\Project-OUT\GRID_02\SUBGRID\DATE
3. **Rename** the folder (need to **add subgrid** in front to avoid confusion)
4. Click **SAVE** button

PROCESS OF STITCHING IMAGE



A Default watermark provided by creator 6. Turn this off because we will use TNB's watermark in Photoshop later.

When this turn on, we will load the car masking with 2 option.

ON left panel :

Go to Imagery > Watermarks

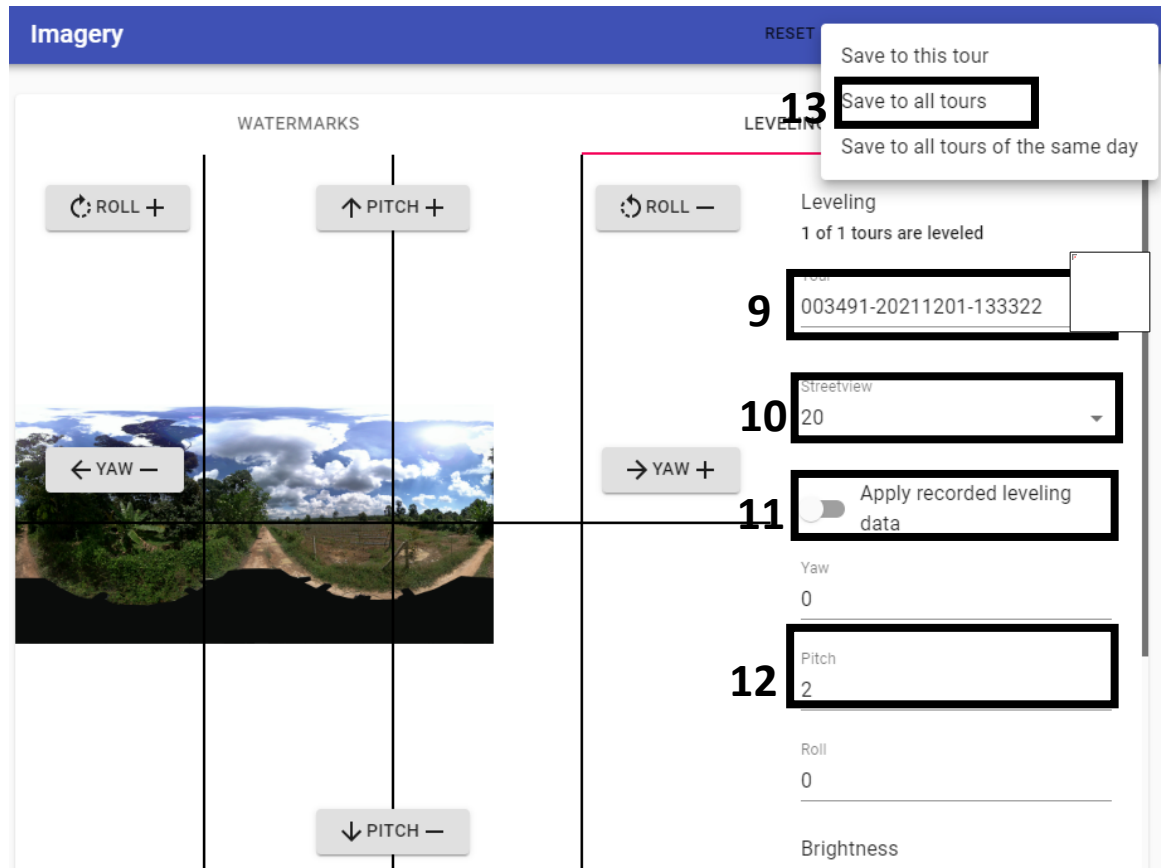
5. **OFF** Add watermark
6. **ON** Add mask
7. Choose **Mask path** (for black masking on the image) – based on the stitching image
8. Click Leveling and brightness to check the images



MMS Mask



Backpack Mask



Go to Imagery > Leveling and Brightness

9. **Current recording tour folder**
10. Current panoramic view
11. **OFF** Applied recorded leveling data
12. Set **pitch** to 2
13. Click **Save** --> and **Save to all tour**

Apply recorded leveling data

Enabled to adjust yaw, pitch and roll

Yaw, Pitch, Roll

Angles between the camera and the leveling sensor.

Brighten shadows

Enabled automatic brightness adjustment for shadows.

Brighten shadows - Strength

Amplifies dark pixels to the threshold level.

Brighten Shadows - Threshold

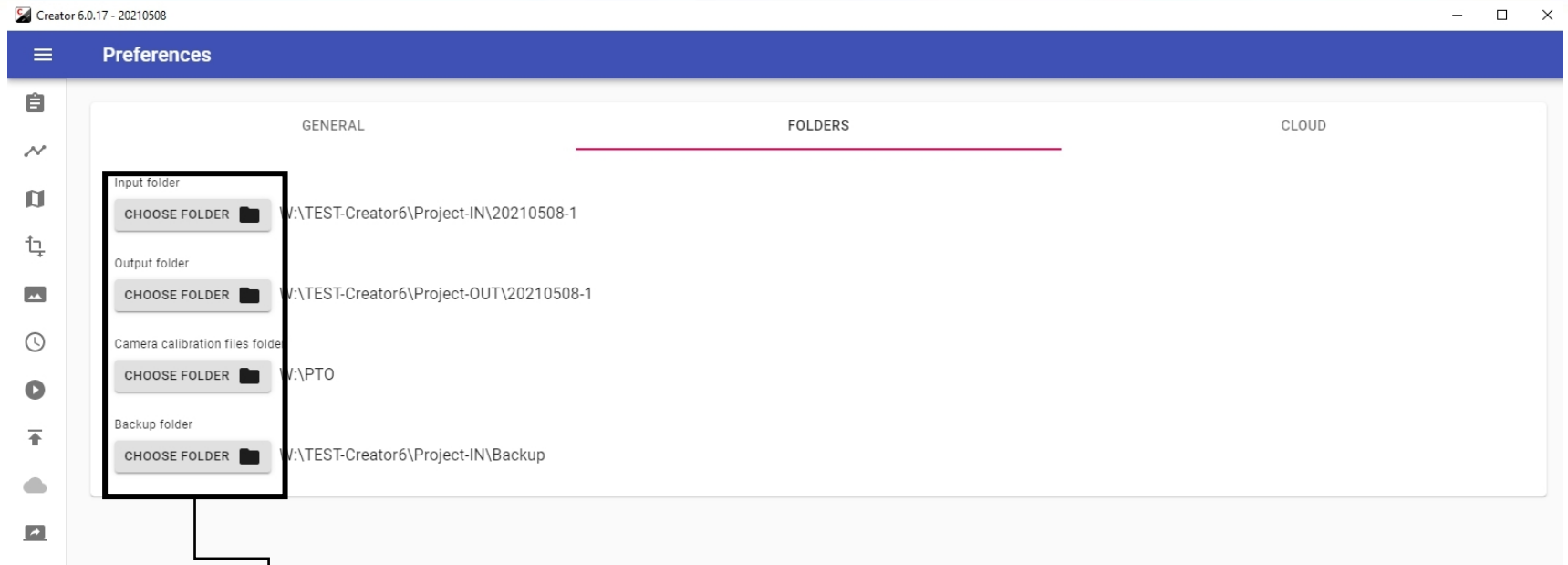
The automatic brightness adjustment will be applied to areas where average brightness is less than the threshold.

Exposure

Adjust image exposure

Go to Preference > Folders

9. Select input folder location
10. Select output folder location
11. Select camera calibration provided from Service provider.
12. Select backup folder location



Input folder - Default location of the recorded projects

Output folder - Default location for publishing. A subfolder with the project name is created automatically.

Camera calibration files folder - Default folder for your camera(s) calibration file(s). They have the .pto and .xml file extension. First Creator 6 looks in the parent folder of a tour. Then in the Camera calibration files folder. In case you have many cameras the right file is picked fully automatic.

Backup folder - Default location for saving project backups.

15

Publish

16

START

Create player

Supports AWS, Microsoft Azure, Google Cloud, Apache, NGINX, IIS or any other web server.

☐

Windows 10 and Windows Server with [Instant Player](#).

Also creates the [Legacy Player](#).

Create streetview-tiles

Required for AWS, Microsoft Azure, Google Cloud, Apache, NGINX, IIS or any other web server.

☒

Optional for Windows 10 and Windows Server with [Instant Player](#).

Create streetviews

For review or 3rd party workflow. Required for [G-Publisher](#) program.

☒

Create project backup

☒

Skip existing streetviews

☒

OPEN OUTPUT FOLDER

☐

Publish to Cloud

Access key ID

Secret access key

Region

Asia Pacific (Singapore) ▼

OPEN CLOUD PLAYER

Go to Publish

15. Follow as picture shown

16. Click START

17. Wait till the process end

Create player - Creates the new player in /Output folder/player2 folder.

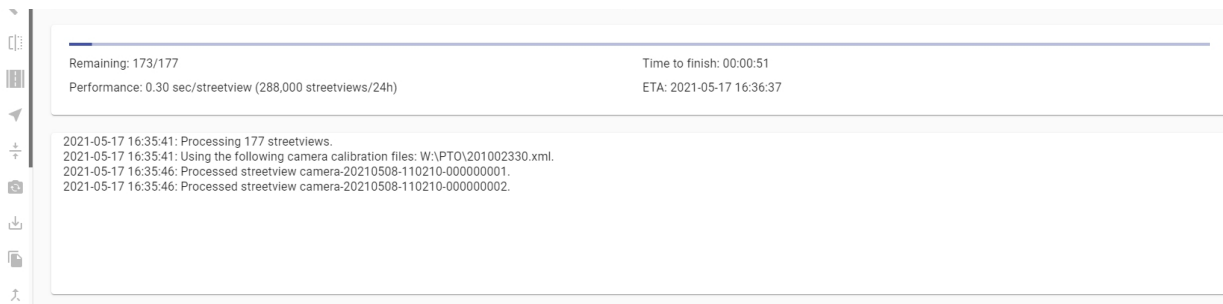
Create streetview-tiles - Create streetview-tiles from source images.

Create streetviews - For G-Publisher, review and a 3rd-party workflow.

Create project backup - Create a new project backup each time publishing is started. The backup file is to be found in the /Output folder/backups folder.

Skip existing streetviews - If the streetview-tiles for a specific streetview already exist, do not process again.

PROCESS STARTED



Remaining: 173/177
Performance: 0.30 sec/streetview (288,000 streetviews/24h)

Time to finish: 00:00:51
ETA: 2021-05-17 16:36:37

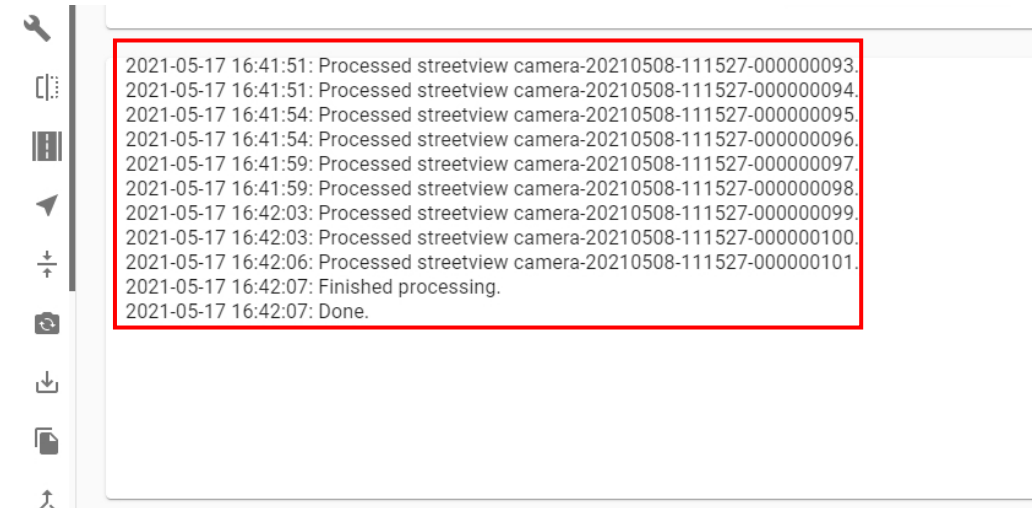
2021-05-17 16:35:41: Processing 177 streetviews.
2021-05-17 16:35:41: Using the following camera calibration files: W:\PTO\201002330.xml.
2021-05-17 16:35:46: Processed streetview camera-20210508-110210-000000001.
2021-05-17 16:35:46: Processed streetview camera-20210508-110210-000000002.

Remaining - remaining image that needs to be processed

Performance - time taken to process per image

Time to finish - remaining time to complete the process

ETA - estimated time



2021-05-17 16:41:51: Processed streetview camera-20210508-111527-000000093.
2021-05-17 16:41:51: Processed streetview camera-20210508-111527-000000094.
2021-05-17 16:41:54: Processed streetview camera-20210508-111527-000000095.
2021-05-17 16:41:54: Processed streetview camera-20210508-111527-000000096.
2021-05-17 16:41:59: Processed streetview camera-20210508-111527-000000097.
2021-05-17 16:41:59: Processed streetview camera-20210508-111527-000000098.
2021-05-17 16:42:03: Processed streetview camera-20210508-111527-000000099.
2021-05-17 16:42:03: Processed streetview camera-20210508-111527-000000100.
2021-05-17 16:42:06: Processed streetview camera-20210508-111527-000000101.
2021-05-17 16:42:07: Finished processing.
2021-05-17 16:42:07: Done.

Details of the completion of the process will be displayed

Output Example

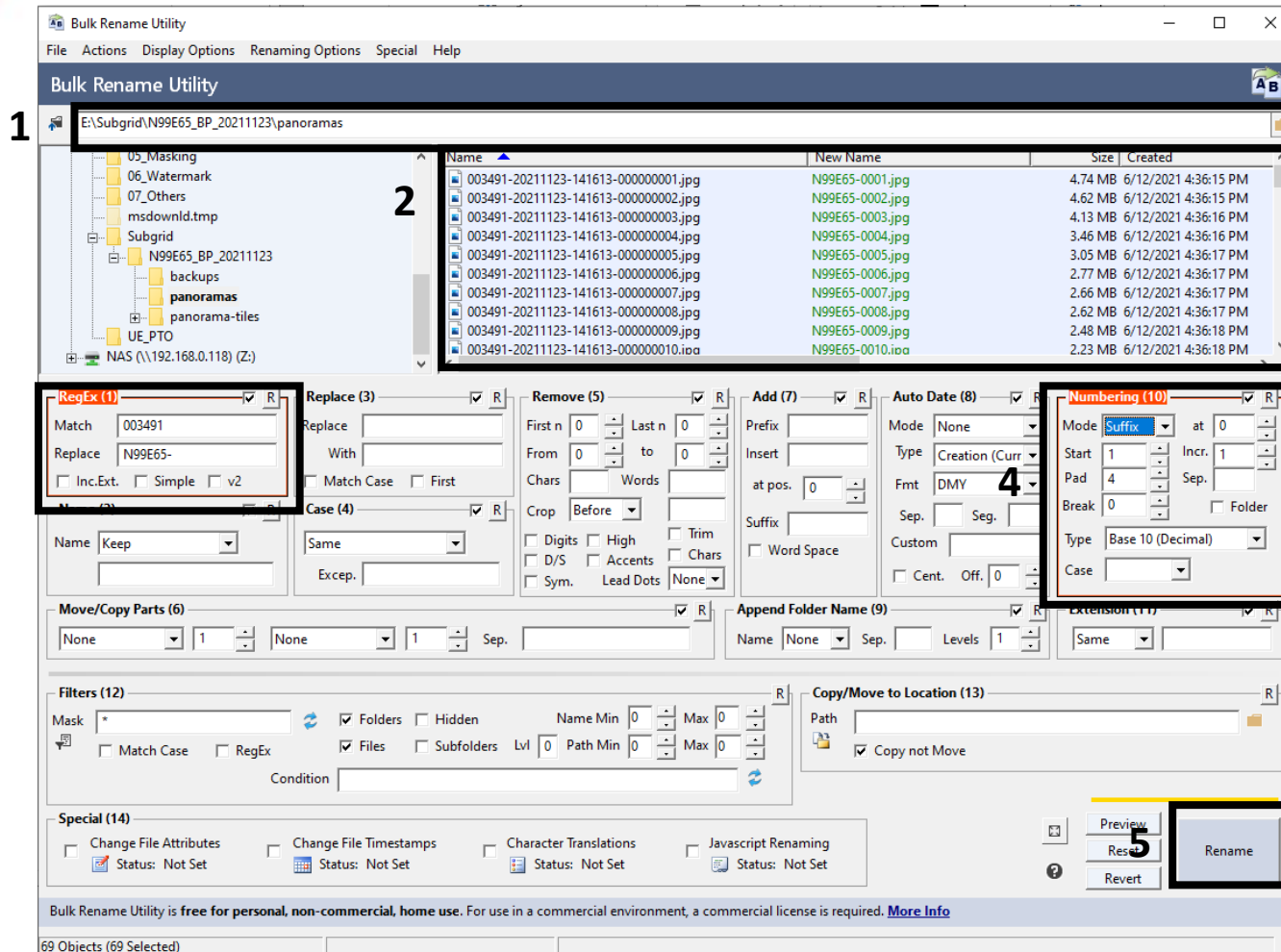
The output of the process will get 2 folders , folder that has a panoramas image and panoramas tiles



This is the last output in process , panorama image with leveling & brightness adjustment and watermark

	Name	Date modified	Type
	panoramas	17/5/2021 4:42 PM	File folder
	panorama-tiles	17/5/2021 4:42 PM	File folder

IMAGE FORMATING & NAMING



Open the Bulk Rename Utility software



1. Make sure the folder directory same as Output folder in Creator **Z:\TNB_LV\03_Creator\Project-OUT NEW\GRID\SUBGRID\DATE\panoramas**
2. Select all output (Ctrl+A)
3. RegEx(1)
 - **Match** – same picture/if different check on window 2 on name which have the repeated number at front.
 - Replace – **SUBGRID-** (DO NOT ADD SPACE AFTER DASH)
4. Numbering(10)
 - **Mode** – Suffix
 - **Start** – Must check the **latest date** on output folder. If non start with 1
 - **Pad** – 4
5. Click **Rename** to start the process
6. Wait till the process is finish

Subgrid-Image Number

Image name before

Image name After

 003491-20220223-135416-000000001
 003491-20220223-135416-000000002
 003491-20220223-135416-000000003
 003491-20220223-135416-000000004
 003491-20220223-135416-000000005
 003491-20220223-135416-000000006
 003491-20220223-135416-000000007
 003491-20220223-135416-000000008
 003491-20220223-135416-000000009
 003491-20220223-135416-000000010
 003491-20220223-135416-000000011
 003491-20220223-135416-000000012
 003491-20220223-135416-000000013
 003491-20220223-135416-000000014
 003491-20220223-135416-000000015
 003491-20220223-135416-000000016
 003491-20220223-135416-000000017
 003491-20220223-135416-000000018
 003491-20220223-135416-000000019
 003491-20220223-135416-000000020
 003491-20220223-135416-000000021
 003491-20220223-135416-000000022
 003491-20220223-135416-000000023

 N93E70-0001.jpg	2.6 ...	JPG File	02/28/2022 16:55:56
 N93E70-0002.jpg	2.5 ...	JPG File	02/28/2022 16:55:56
 N93E70-0003.jpg	2.4 ...	JPG File	02/28/2022 16:55:57
 N93E70-0004.jpg	2.4 ...	JPG File	02/28/2022 16:55:57
 N93E70-0005.jpg	2.4 ...	JPG File	02/28/2022 16:55:59
 N93E70-0006.jpg	2.4 ...	JPG File	02/28/2022 16:55:59
 N93E70-0007.jpg	2.4 ...	JPG File	02/28/2022 16:56:01
 N93E70-0008.jpg	2.3 ...	JPG File	02/28/2022 16:56:01
 N93E70-0009.jpg	2.3 ...	JPG File	02/28/2022 16:56:03
 N93E70-0010.jpg	2.2 ...	JPG File	02/28/2022 16:56:03
 N93E70-0011.jpg	2.2 ...	JPG File	02/28/2022 16:56:05
 N93E70-0012.jpg	2.2 ...	JPG File	02/28/2022 16:56:05
 N93E70-0013.jpg	2.2 ...	JPG File	02/28/2022 16:56:07
 N93E70-0014.jpg	2.2 ...	JPG File	02/28/2022 16:56:07
 N93E70-0015.jpg	2.2 ...	JPG File	02/28/2022 16:56:09
 N93E70-0016.jpg	2.1 ...	JPG File	02/28/2022 16:56:09
 N93E70-0017.jpg	2.1 ...	JPG File	02/28/2022 16:56:11
 N93E70-0018.jpg	2.1 ...	JPG File	02/28/2022 16:56:11
 N93E70-0019.jpg	2.1 ...	JPG File	02/28/2022 16:56:13
 N93E70-0020.jpg	2.1 ...	JPG File	02/28/2022 16:56:13
 N93E70-0021.jpg	2.1 ...	JPG File	02/28/2022 16:56:14
 N93E70-0022.jpg	2.1 ...	JPG File	02/28/2022 16:56:14
 N93E70-0023.jpg	2.1 ...	JPG File	02/28/2022 16:56:16

IMAGE ENHANCEMENT USING ADOBE LIGHTROOM



CREATING A PRESET - ADOBE LIGHTROOM

Go to Develop at the right side panel



Go to Tone adjustment

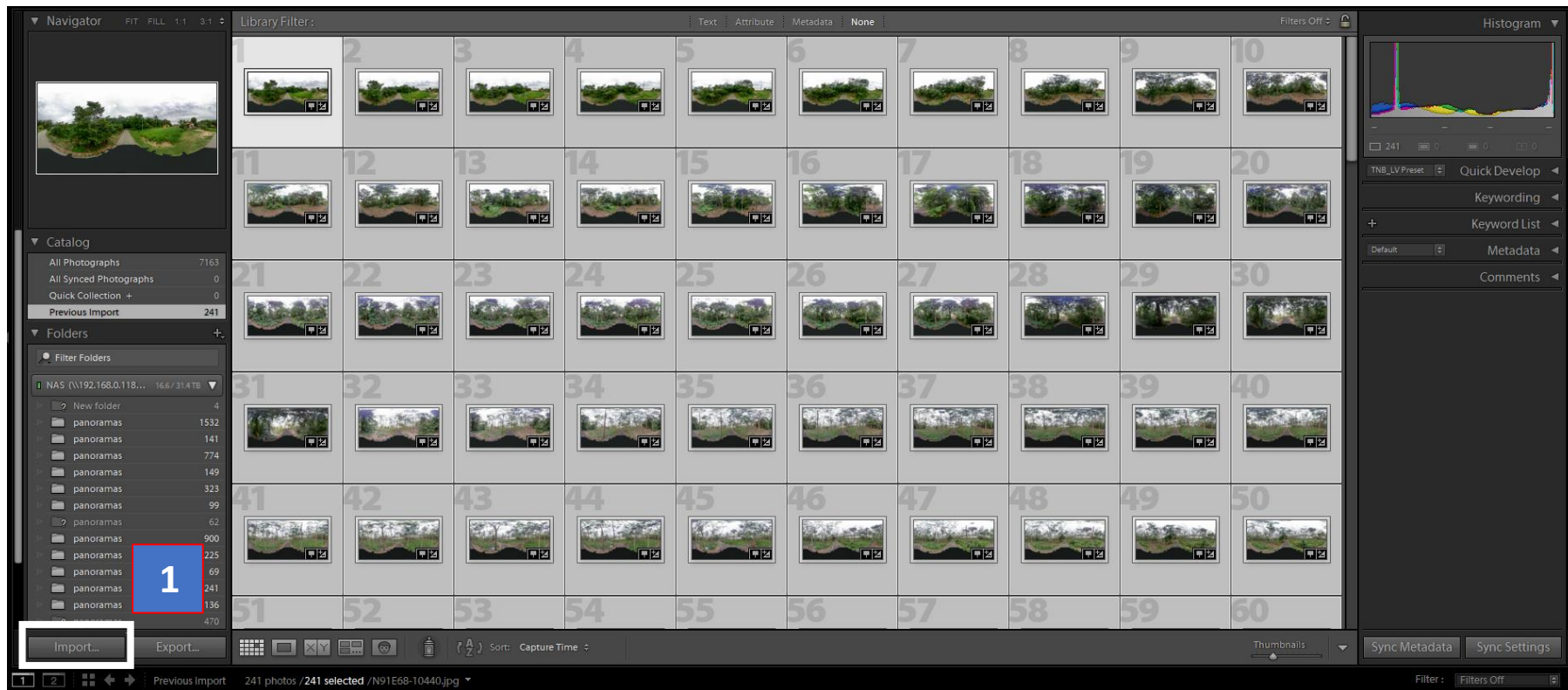
1. Exposure +0.25
2. Shadow +60
3. Whites -65
4. Blacks +30

Go to Presence:

1. Clarity +24
2. Vibrance +10

Basic setting for image Enhancement and it does not affect the picture Quality.

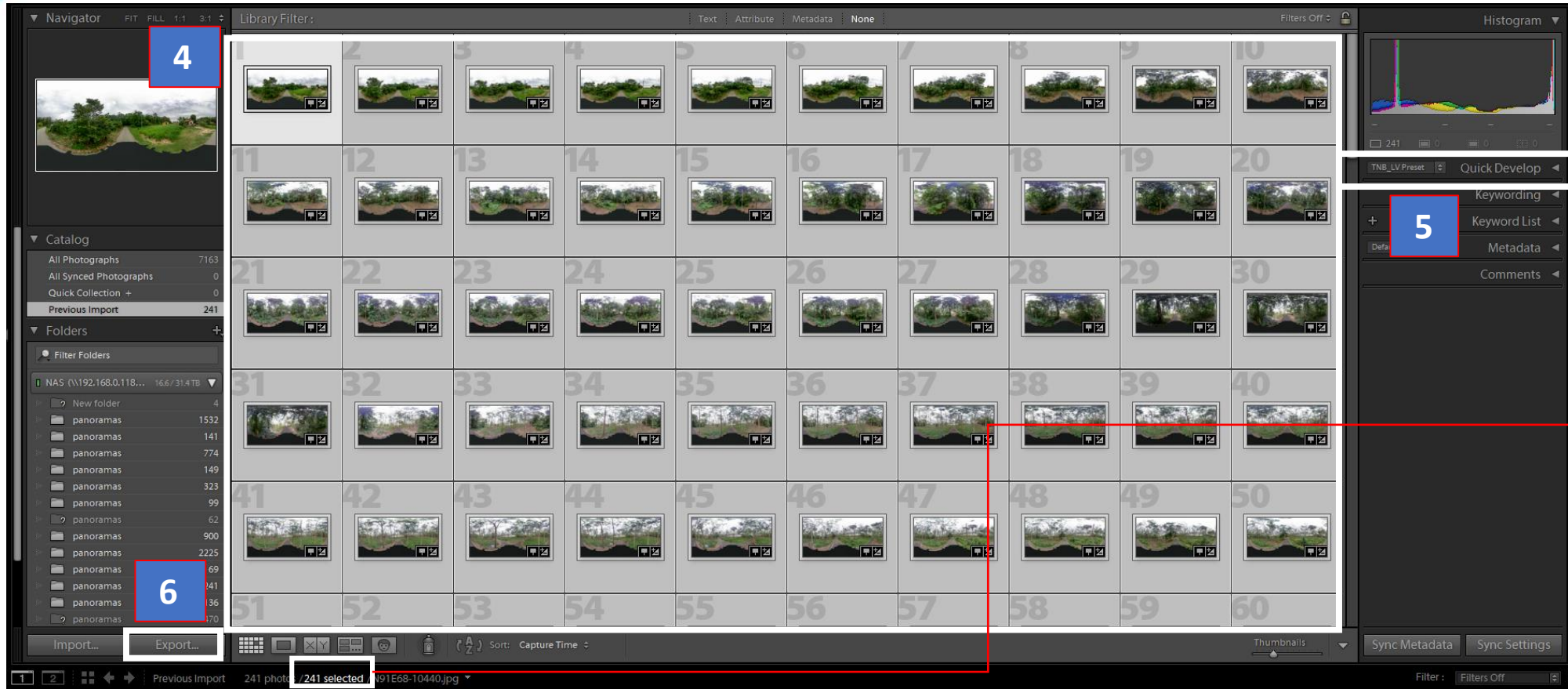
APPLYING A PRESET



1. On the left panel, select import from folder location

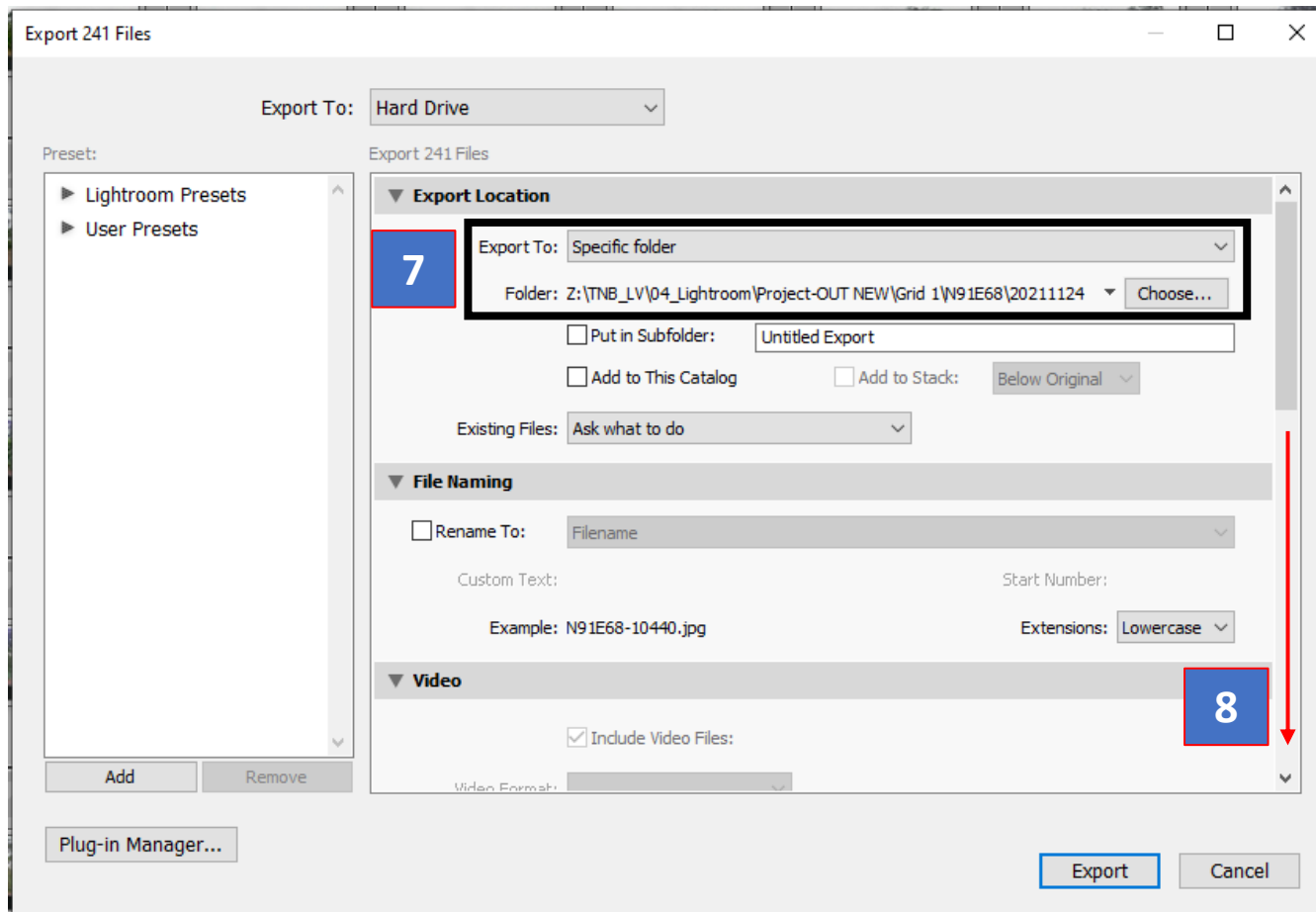


2. Select the Source (make sure the input is from Creator Output with the rename image)
**Z:\TNB_LV\03_Creator\Project-OUT
NEW\GRID\SUBGRID\DATE
\panoramas**
3. Select Import to proceed



Total image number

4. Wait for all images have been upload then **select one image and select Ctrl+A to select all images**
5. Go to **Quick Develop>User Presets>TNB_LV Presets** and wait till the presets apply to all the images
6. Select Export

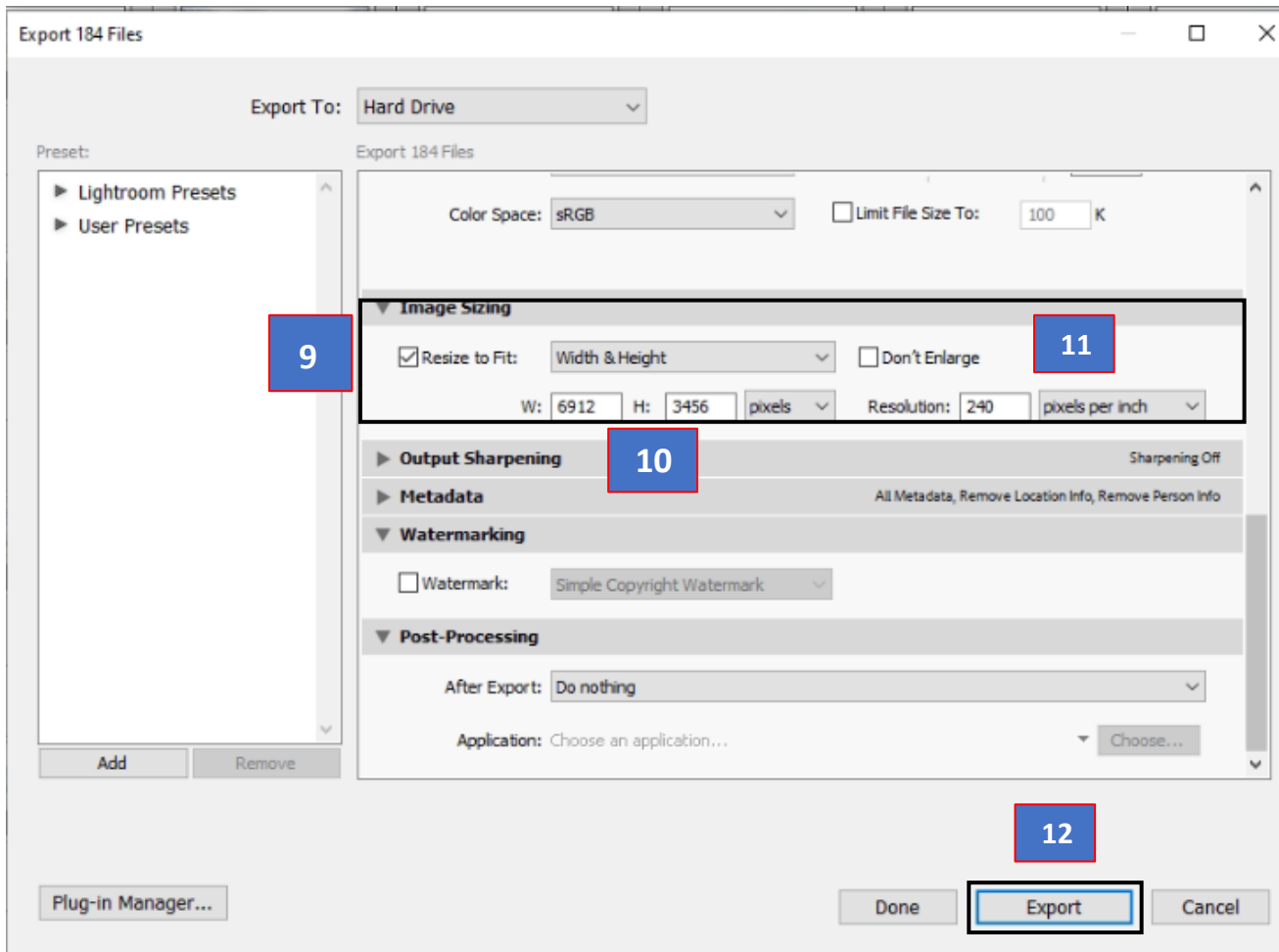


7. Make sure :

- Export To – Specific folder and click Choose.
- The folder directory -
**Z:\TNB_LV\04_Lightroom\Project-OUT
NEW\GRID\SUBGRID\DATE**

8. Scroll down to image sizing ↓

IMAGE EXPORT



9. Tick Resize to fit
10. Set width & height 6912 x 3456
11. Resolution set to 240 & pixel per inch
12. Click export and wait till process finish

Minimum specifications image dimension

No.	Items	Specifications
1.	Camera System	Minimum 30 MP, 360° Spherical Camera Imaging System
2.	Image Format	Equirectangular , .jpg
3.	Image Resolution & Maximum Image Size	Minimum resolution : 30 MP (5MP x 6 sensors) Image Size : Less than 10 MB per image
4.	Image Dimension	Minimum 6912 x 3456 in pixel format (Image ratio 2:1)

OUTPUT EXAMPLE



Before Enhancement

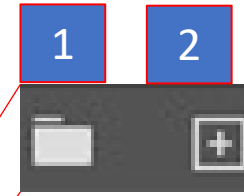


After Enhancement

IMAGE MASKING USING ADOBE PHOTOSHOP

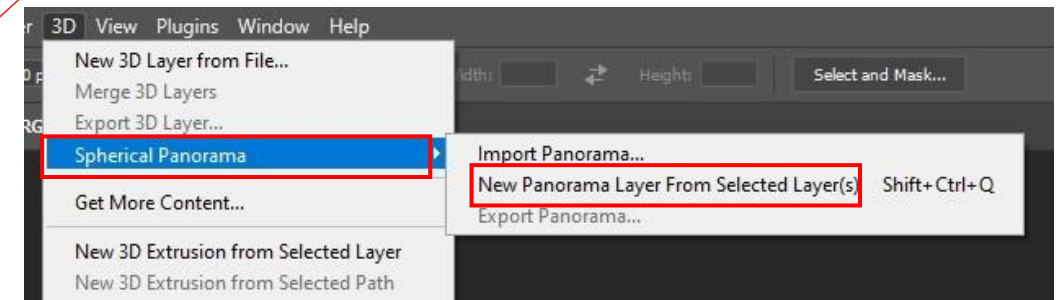


ADOBE PHOTOSHOP – CREATING A BATCH PROCESSING



Process to make batch processing

1. Create a new action folder
2. Click [+] to create new action. Once created the action will be recorded



Convert the panorama into 3D image by going to 3D > Spherical panorama and selecting “New Panorama Layer From Selected Layer.” This will convert your workspace in to a 3D Workspace

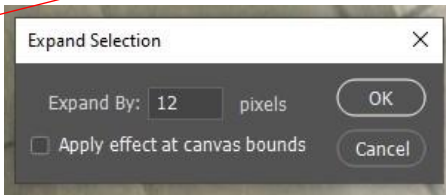
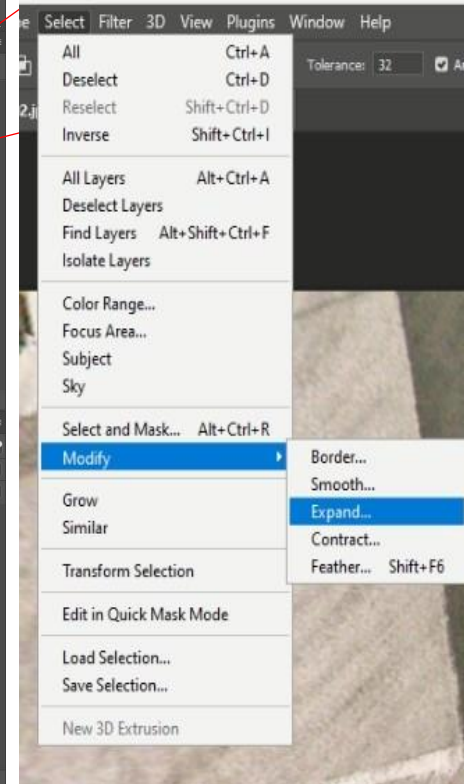
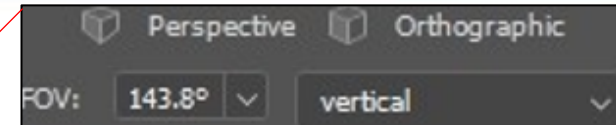
MASKING & WATERMARK PROCESS

5. Hold left mouse button and drag it all the way down. The view should be as in picture shown

6. Under properties, change the FOV to vertical and adjust the slider until the value is between 140- 145

7. Select the magic wand tool and then select the car

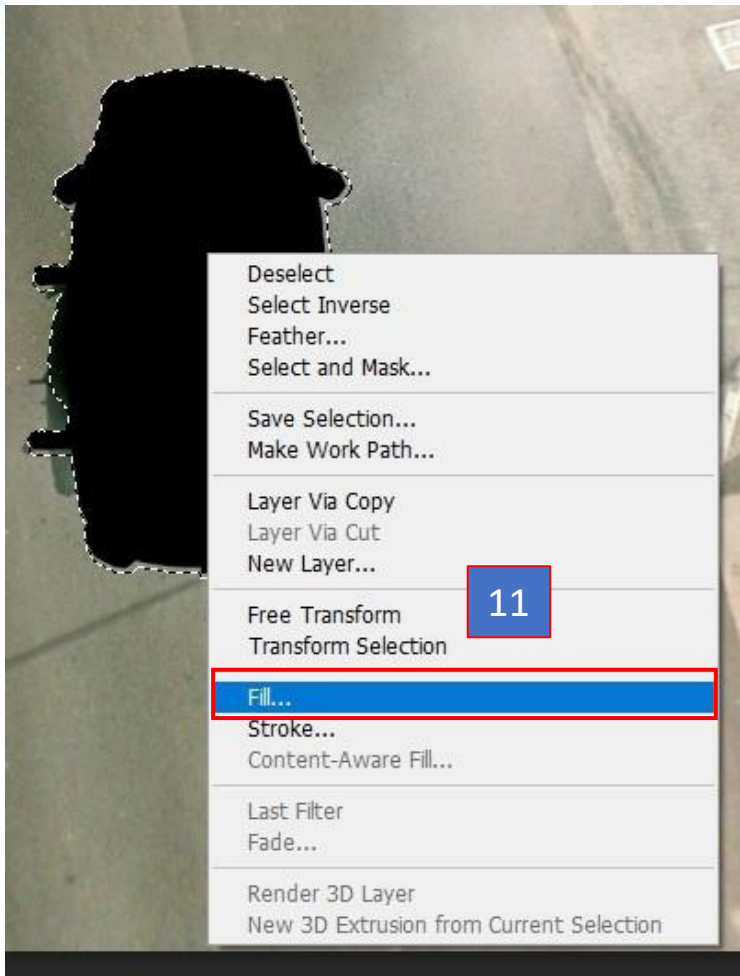
8. Once you have selected the car, it will look like this



9. Go to select > Modify > Expand

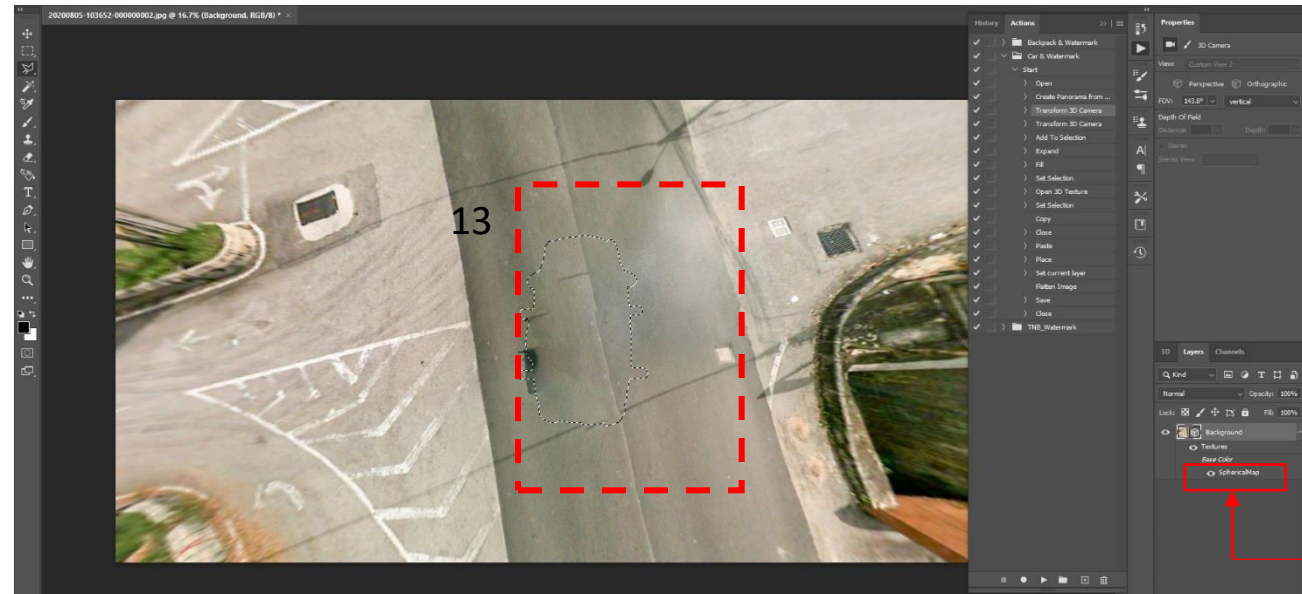
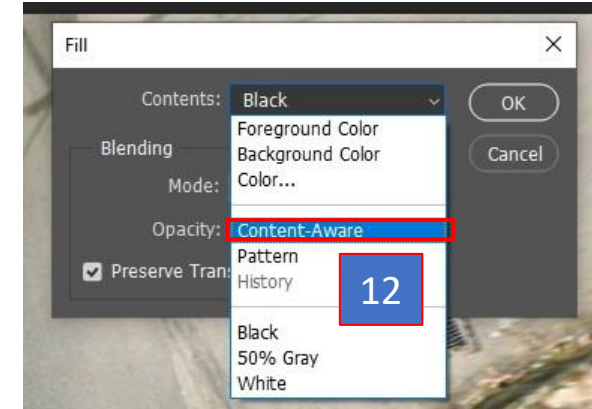
10. Add a value between 10 - 15 pixels and click OK, the selection should be expanded (refer image below)

MASKING & WATERMARK PROCESS



11. Select the polygonal lasso tool, right click your selection and choose fill

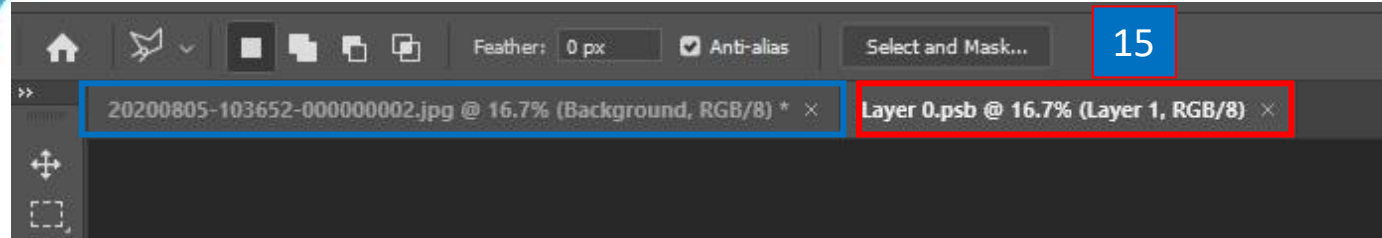
12. Click the content options and choose Content-Aware. Then click OK



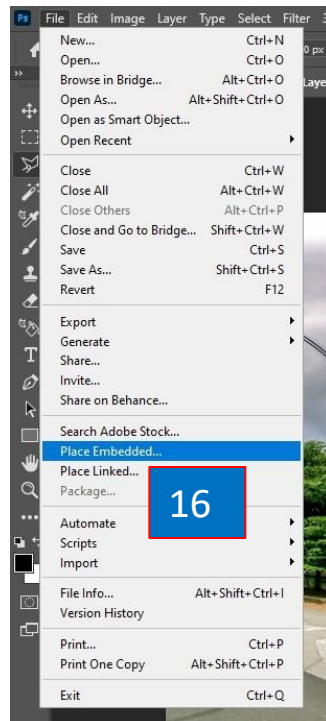
13. Once Content-Aware fill has removed the object in the selection, press ctrl + D to remove the selection

14. Convert the image back to a panorama. Go to layers and double click SphericalMap

MASKING & WATERMARK PROCESS

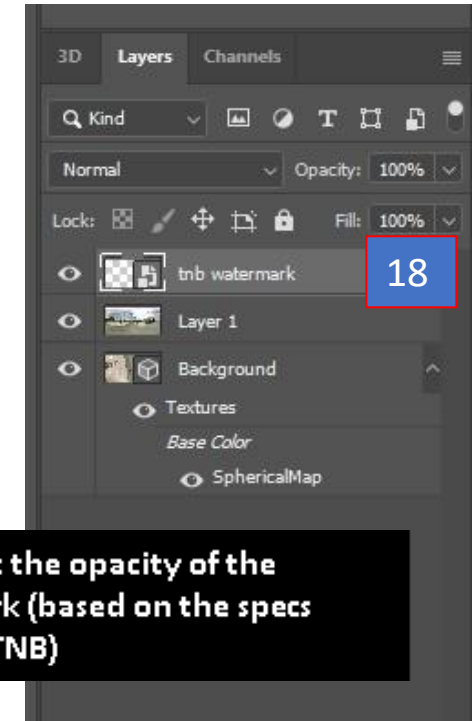
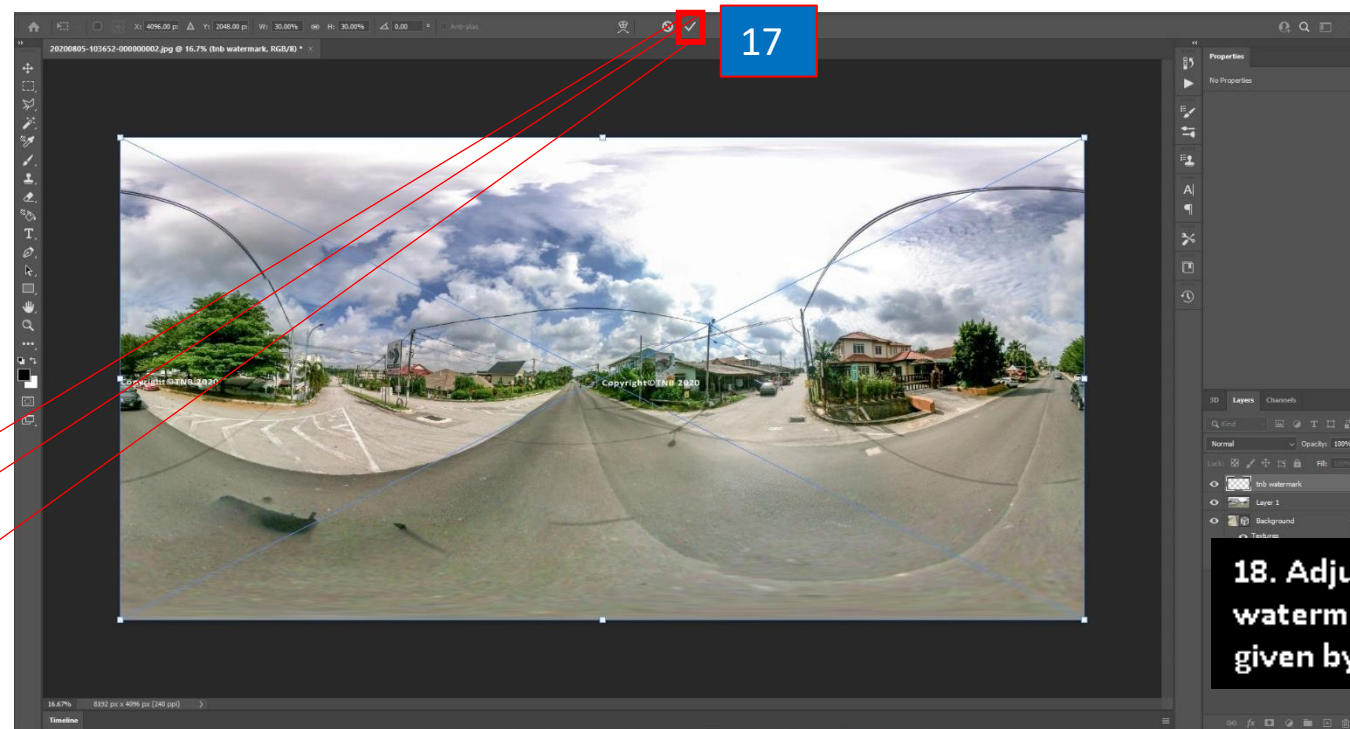


15. This will open a new canvas. Ctrl + A to select the pano and Ctrl + X to cut the selected pano. Close this canvas (highlighted in red), go back to the previous canvas (highlighted in blue) and Ctrl + Shift + V to paste the image

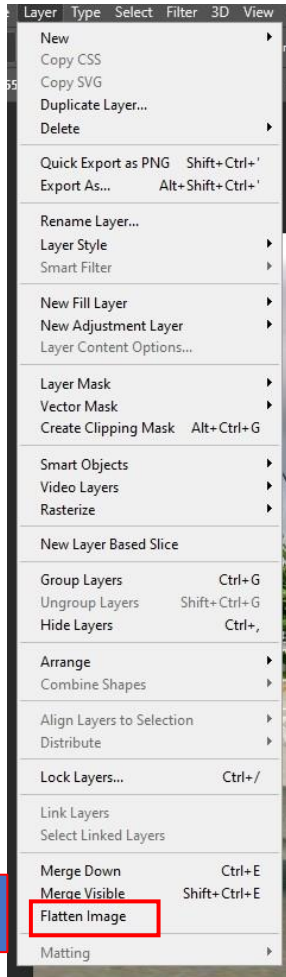


16. Now we are going to add a watermark to the pano. Go to file > place embedded and select the watermark image that you want to use

17. Click on the check mark to apply the watermark image

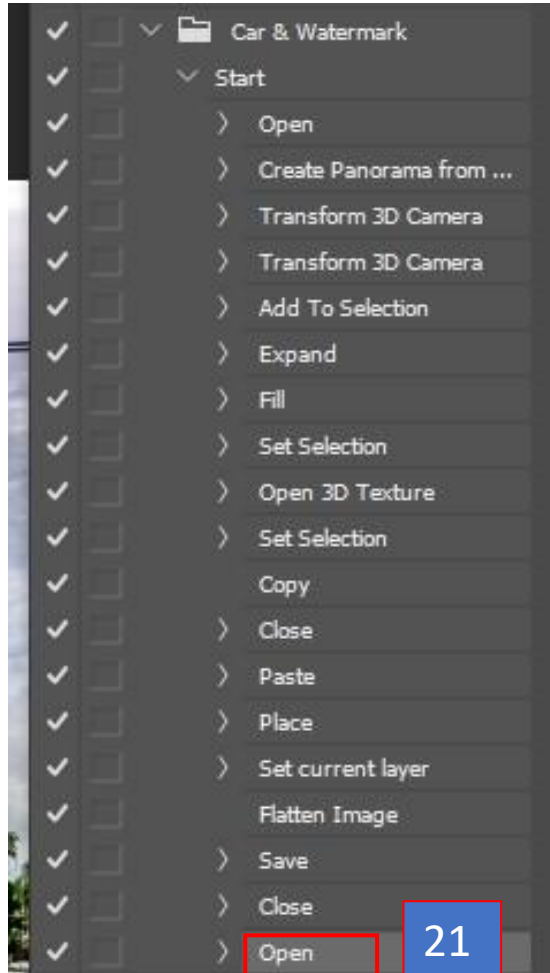


18. Adjust the opacity of the watermark (based on the specs given by TNB)



19. Go to layer > Flatten image to merge the layers together

20. A save action must be created, do this by going to File > Save As. The naming convention and location of where the pano is saved doesn't matter as this step is to only record the save action.

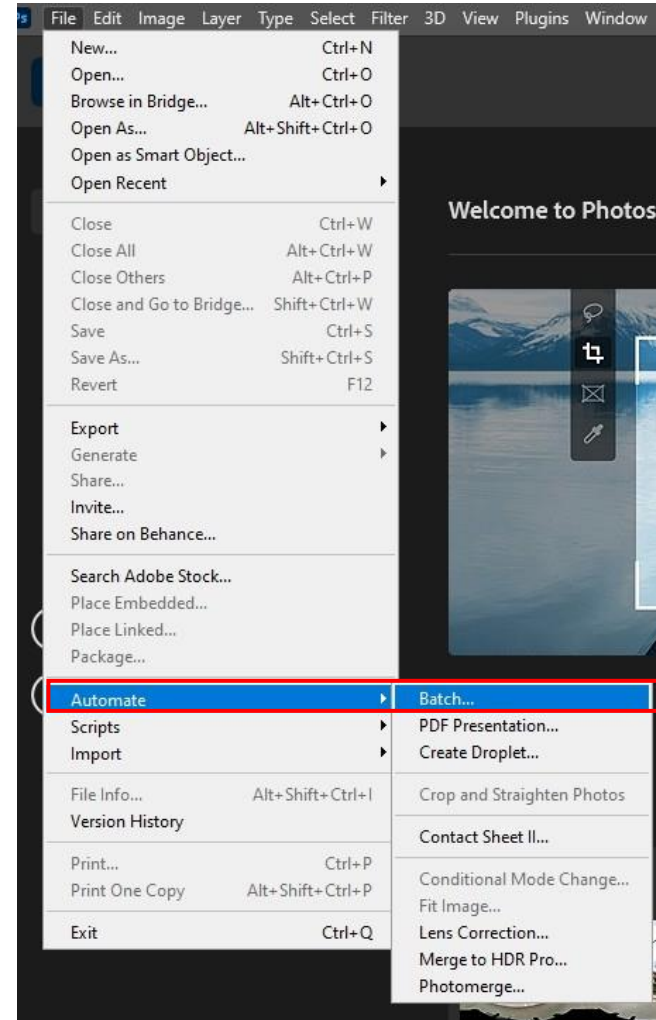


21. Close the canvas to record the close action sequence. Re-open the canvas again to stop the recording. Make sure to delete the extra open sequence at the bottom, as your last action sequence in the folder should be the close sequence.

BATCH AUTOMATE PROCESSING

1. To begin the batch processing, go to **File > Automate > Batch**

1



MASKING & WATERMARK PROCESS

2. Select the action folder

3. For input, Choose “folder” as the source and select input location. Make sure to check all on the boxes to prevent from pop ups when process start

4. Select “Log Errors to File” and then save the file in a designated logfolder. Any errors that occur during the batch processing will be recorded in the logs

5. Now we are going to set the output settings. Choose “folder” as the source and select the output location. Make sure the box is checked off to prevent pop ups

6. This step is to set the naming convention of the panos. In the first box, choose “Document Name.” This will retain the name of the panos. In the second box, choose “extension”

7. Once all the steps above have been completed, click OK to begin the batch processing

The screenshot shows the 'Batch' processing window with the following settings:

- 2** Play Set: Car & Watermark Action: Start
- 3** Source: Folder Choose... W:\01_TNB_LV\...\N87E40\02_Backpack\20210508\
 - ☒ Override Action "Open" Commands
 - ☒ Include All Subfolders
 - ☒ Suppress File Open Options Dialogs
 - ☒ Suppress Color Profile Warnings
- 4** Errors: Log Errors to File Save As... W:\01_TNB_LV\...\LOG\N87E39-Backpack-LOG-8.txt
- 5** Destination: Folder Choose... W:\01_TNB_LV\03_PROCESSING\...\N87E40\02_Backpack\20210508\
 - ☒ Override Action "Save As" Commands
- 6** File Naming Example: MyFile.gif

Document Name	Document Name
extension	extension
	None
	None
	None
	None
- Starting Serial #: 1
- Compatibility: ☒ Windows ☐ Mac OS ☐ Unix

OUTPUT EXAMPLE



PROCESSING PERFORMANCE

Performance tracking based on daily process

	RTX 3050	RTX 2060	RTX 3050	RTX 3050
	PRIVACY KEEPER	CREATOR	LIGHTROOM	PHOTOSHOP
Time per image	0.24 sec	0.30 sec	1.2 sec	7.6 sec
0-100 images	24 sec	30 sec	2 min 30 sec	12 min 30 sec
0-1000 images	2 – 3 min	3 – 4 min	20 – 25 min	1 hour 30 min
0-10000 images	20 – 30 min	30 – 40 min	3 – 4 hours	21 – 22 hours

However, the performance might be different on the part of GPU installed. Higher GPU will taking a less time to process, while the lower GPU will affected the process performance. Use a higher GPU is recommended 

METADATA EDITING & FORMATING



2CSV APP & EXCEL

Input folder

C:\Users\USER\Desktop\PROCESSING TOTK\Data Processing\02_Privacy Keeper\20220904

C:\Users\USER\Desktop\PROCESSING TOTK\Data Processing\02_Privacy Keeper\20220904

Name	Date modified	Type
003485-20220904-144310	3/5/2024 10:25 AM	File folder
003485-20220904-144428	3/5/2024 10:25 AM	File folder
003485-20220904-144543	3/5/2024 10:25 AM	File folder
003485-20220904-144647	3/5/2024 10:25 AM	File folder
003485-20220904-145542	3/5/2024 10:25 AM	File folder

- In this process, we will sort the panoramas metadata and place it to output folder (metadata folder) with a date provided from MMS/BP tour:

- Select the input folder location as highlighted above
- Open 2csv_apps → copy panorama filename
- Select output folder to save the result

URBAN
EXPLORER
Maker of Digital Twins

Output folder

Home Page

The path for the CSV files contains:

C:\Users\USER\Desktop\PROCESSING TOTK\Data Processing\03_Creator\Project -OUT\Grid_01\N93E70\20220904

Note: The program deals ONLY with panoramas.csv file.

The path to save the results:

C:\Users\USER\Desktop\PROCESSING TOTK\Data Processing\01_Metadata\GRID_1\N93E70

Submit

Shutdown the service

Name	Date modified	Type	Size
003485-20220904-144310	10/5/2024 3:34 PM	Microsoft Excel Comma...	6 KB
003485-20220904-144428	10/5/2024 3:34 PM	Microsoft Excel Comma...	6 KB
003485-20220904-144543	10/5/2024 3:34 PM	Microsoft Excel Comma...	5 KB
003485-20220904-144647	10/5/2024 3:34 PM	Microsoft Excel Comma...	6 KB
003485-20220904-145542	10/5/2024 3:34 PM	Microsoft Excel Comma...	7 KB

A default metadata filename is "panorama.csv" the filename should be remain the same as the tour name for a record and avoid confusion.

Seccussfully!

file has been renamed successfully

Home Page

The path for the CSV files contains:

C:\Users\USER\Desktop\PROCESSING TOTK\Data Processing\01_Metadata\GRID_1\N93E70

* Note: The program deals ONLY with csv file.

Submit

Shutdown the service

C:\Users\USER\Desktop\PROCESSING TOTK\Data Processing\01_Metadata\GRID_1\N93E70

Name	Date modified	Type	Size
003485-20220904-144310	10/5/2024 3:34 PM	Microsoft Excel Comma...	
003485-20220904-144428	10/5/2024 3:34 PM	Microsoft Excel Comma...	

The file has been merged into one metadata from a same date tour

Output File location directory

- Next, all metadata is ready to merge into one single metadata
 - Select the metadata file to merge
 - Open 2csv_apps → merge metadata
 - Select output folder to save the result

Name	Date modified	Type	Size
003485-20220904-144310	10/5/2024 3:37 PM	Microsoft Excel Comma...	6 KB
003485-20220904-144428	10/5/2024 3:37 PM	Microsoft Excel Comma...	7 KB
003485-20220904-144543	10/5/2024 3:37 PM	Microsoft Excel Comma...	6 KB
003485-20220904-144647	10/5/2024 3:37 PM	Microsoft Excel Comma...	7 KB
003485-20220904-145542	10/5/2024 3:37 PM	Microsoft Excel Comma...	7 KB
20220904	10/5/2024 3:37 PM	Microsoft Excel Comma...	28 KB

METADATA FORMAT

filename	latitude	longitude	heading	pitch	roll	date	time
This filename on csv must same as filename in <u>pang</u> image.	At least 6 decimal places.	At least 6 decimal places.	Unit in degree. Only 'heading' is accepted. 'direction' and 'yaw' will be rejected.	Unit in degree.	Unit in degree.	In format (YYYY/MM/DD)	In format (HH:MM:SS), 24-hour format
000000.654.jpg	2.3264565	101.235648	125.6548 or 104.2158	8.2613	1.0365 or -1.5463	2019/08/13	16:52:31

FINAL OUTPUT

filename	latitude	longitude	roll	pitch	heading	date	time
N93E70-0001.jpg	2.542421	102.8077	0	0	255.8094	4/9/2022	43:00.4
N93E70-0002.jpg	2.542429	102.8078	0	0	255.7978	4/9/2022	43:01.0
N93E70-0003.jpg	2.542437	102.8078	0	0	254.6215	4/9/2022	43:02.5
N93E70-0004.jpg	2.542449	102.8078	0	0	239.9194	4/9/2022	43:04.1
N93E70-0005.jpg	2.542471	102.8078	0	0	209.8567	4/9/2022	43:05.7
N93E70-0006.jpg	2.542498	102.8078	0	0	174.5648	4/9/2022	43:07.4
N93E70-0007.jpg	2.542524	102.8078	0	0	171.2146	4/9/2022	43:08.2
N93E70-0008.jpg	2.542552	102.8078	0	0	176.4449	4/9/2022	43:09.1
N93E70-0009.jpg	2.54258	102.8078	0	0	179.6768	4/9/2022	43:10.2
N93E70-0010.jpg	2.542608	102.8078	0	0	181.3509	4/9/2022	43:11.2
N93E70-0011.jpg	2.542638	102.8078	0	0	183.5656	4/9/2022	43:12.0
N93E70-0012.jpg	2.542666	102.8078	0	0	185.6539	4/9/2022	43:12.9
N93E70-0013.jpg	2.542693	102.8078	0	0	184.7171	4/9/2022	43:13.6
N93E70-0014.jpg	2.54272	102.8078	0	0	181.4342	4/9/2022	43:14.3
N93E70-0015.jpg	2.542747	102.8078	0	0	177.9373	4/9/2022	43:16.0

Filename
should be
synchronize
with image

Latlong coordinate
extracted from camera

DATA PUBLISHING TO WEBPLAYER

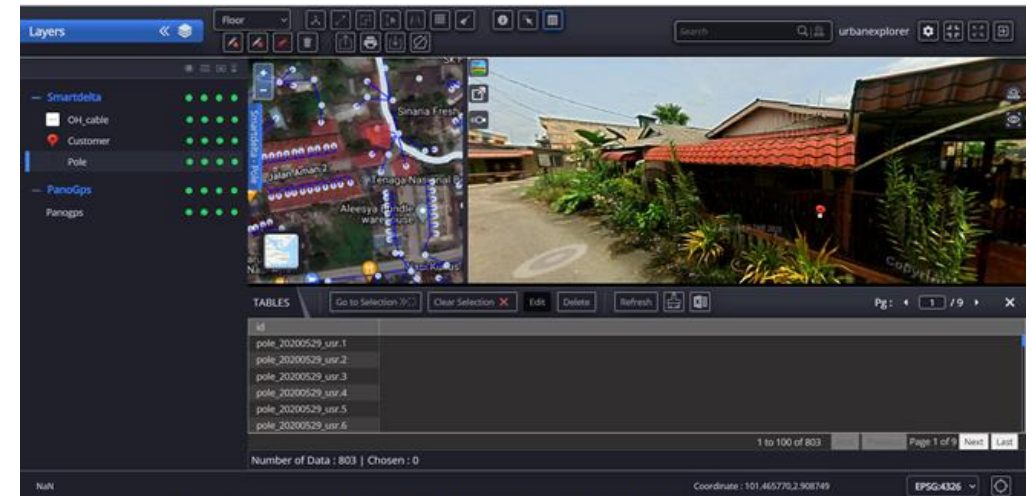
DATA PREPARATION



CLOUD STORAGE



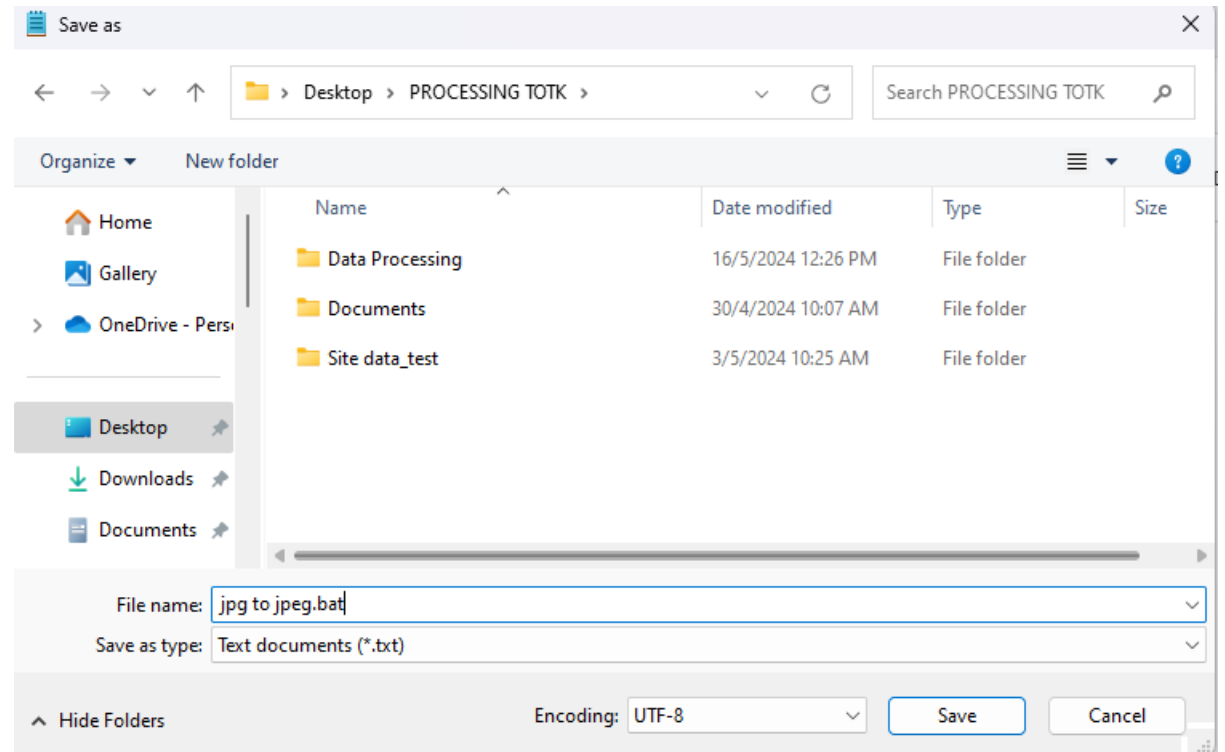
USV WEBPLAYER



URBAN STREETVIEW STANDARD INTERFACE

CONVERT – JPG TO JPEG

1. Open notepad to create batch file.
2. Type `ren *.jpg *.jpeg` and save file as .bat
3. Open .jpg folder location example
**Z:\TNB_LV\09_PUBLISH\Project-
IN\GRID\SUBGRID\DATE** and paste the batch file
to start the process.
3. Wait until process finish.



CONVERT – JPG TO SRF

1. In linux terminal window, type “srfexporter” in the line

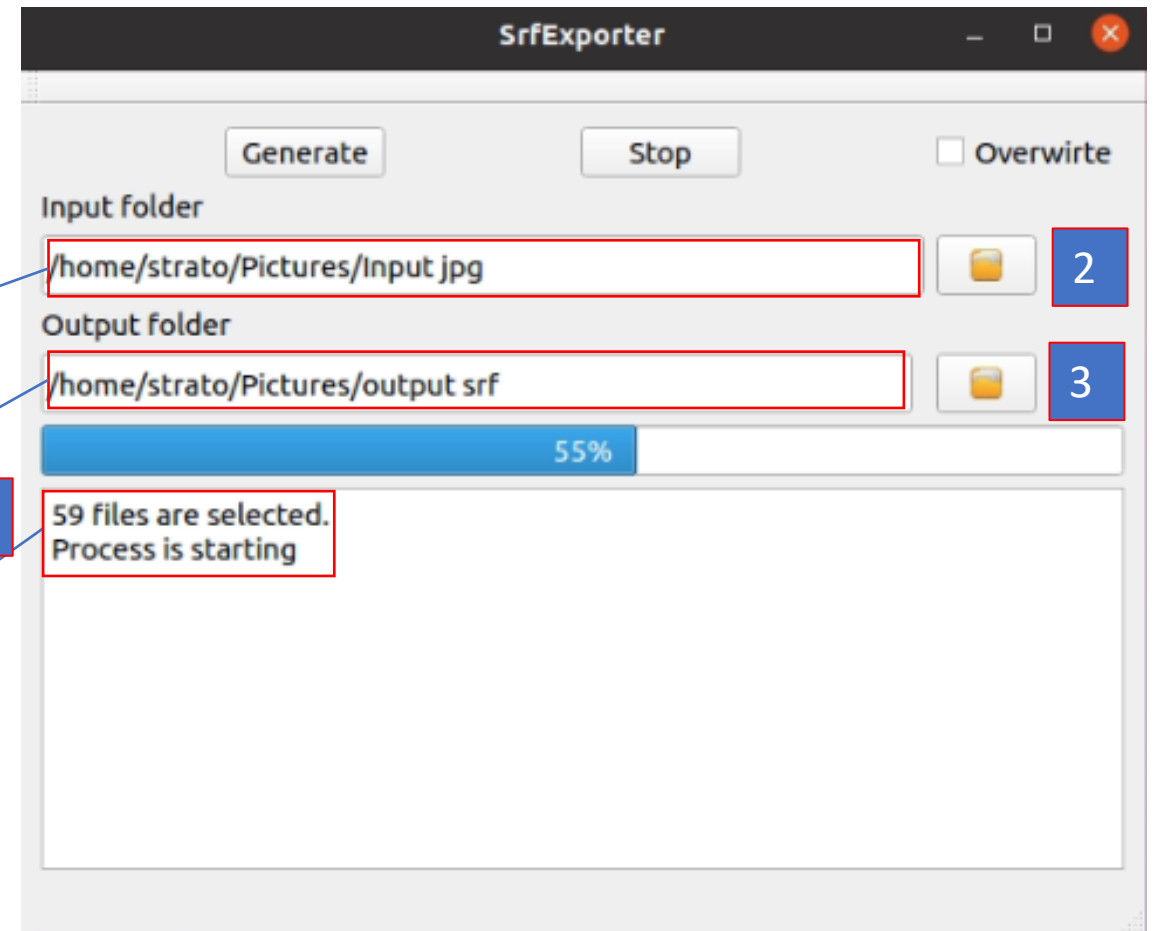
1

```
strato@strato-proc: ~  
(base) strato@strato-proc:~$ srfexporter
```

2. Input folder location :
Navigate to the location of the
final .jpg image

3. Output folder location : the
file where should be save

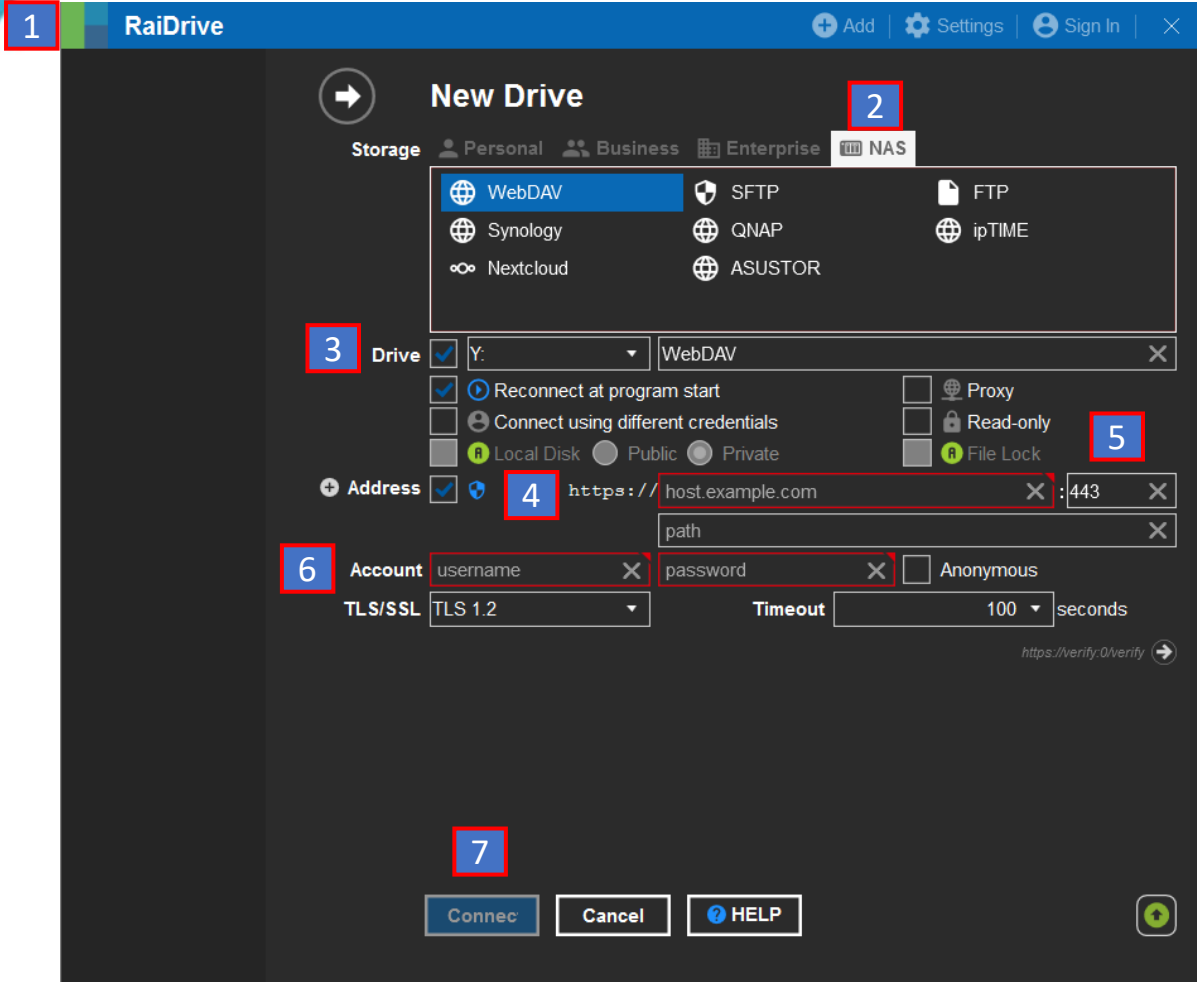
4.Total selected files in folder
And a status of image
processing



The images are ready for data publishing

 N93E70-0001	16/5/2024 12:26 PM	JPEG File
 N93E70-0002	16/5/2024 12:26 PM	JPEG File
 N93E70-0003	16/5/2024 12:26 PM	JPEG File
 N93E70-0004	16/5/2024 12:26 PM	JPEG File
 N93E70-0005	16/5/2024 12:26 PM	JPEG File
 N93E70-0006	16/5/2024 12:26 PM	JPEG File
 N93E70-0007	16/5/2024 12:26 PM	JPEG File
 N93E70-0008	16/5/2024 12:26 PM	JPEG File
 N93E70-0009	16/5/2024 12:26 PM	JPEG File
 N93E70-0010	16/5/2024 12:26 PM	JPEG File
 N93E70-0011	16/5/2024 12:26 PM	JPEG File
 N93E70-0012	16/5/2024 12:26 PM	JPEG File
 N93E70-0013	16/5/2024 12:26 PM	JPEG File
 N93E70-0014	16/5/2024 12:26 PM	JPEG File
 N93E70-0015	16/5/2024 12:26 PM	JPEG File
 N93E70-0016	16/5/2024 12:26 PM	JPEG File

 N93E70-0001.srf	16/5/2024 2:24 PM	SRF File
 N93E70-0002.srf	16/5/2024 2:24 PM	SRF File
 N93E70-0003.srf	16/5/2024 2:24 PM	SRF File
 N93E70-0004.srf	16/5/2024 2:24 PM	SRF File
 N93E70-0005.srf	16/5/2024 2:24 PM	SRF File
 N93E70-0006.srf	16/5/2024 2:24 PM	SRF File
 N93E70-0007.srf	16/5/2024 2:24 PM	SRF File
 N93E70-0008.srf	16/5/2024 2:24 PM	SRF File
 N93E70-0009.srf	16/5/2024 2:24 PM	SRF File
 N93E70-0010.srf	16/5/2024 2:24 PM	SRF File
 N93E70-0011.srf	16/5/2024 2:24 PM	SRF File
 N93E70-0012.srf	16/5/2024 2:24 PM	SRF File
 N93E70-0013.srf	16/5/2024 2:24 PM	SRF File
 N93E70-0014.srf	16/5/2024 2:24 PM	SRF File
 N93E70-0015.srf	16/5/2024 2:24 PM	SRF File
 N93E70-0016.srf	16/5/2024 2:24 PM	SRF File



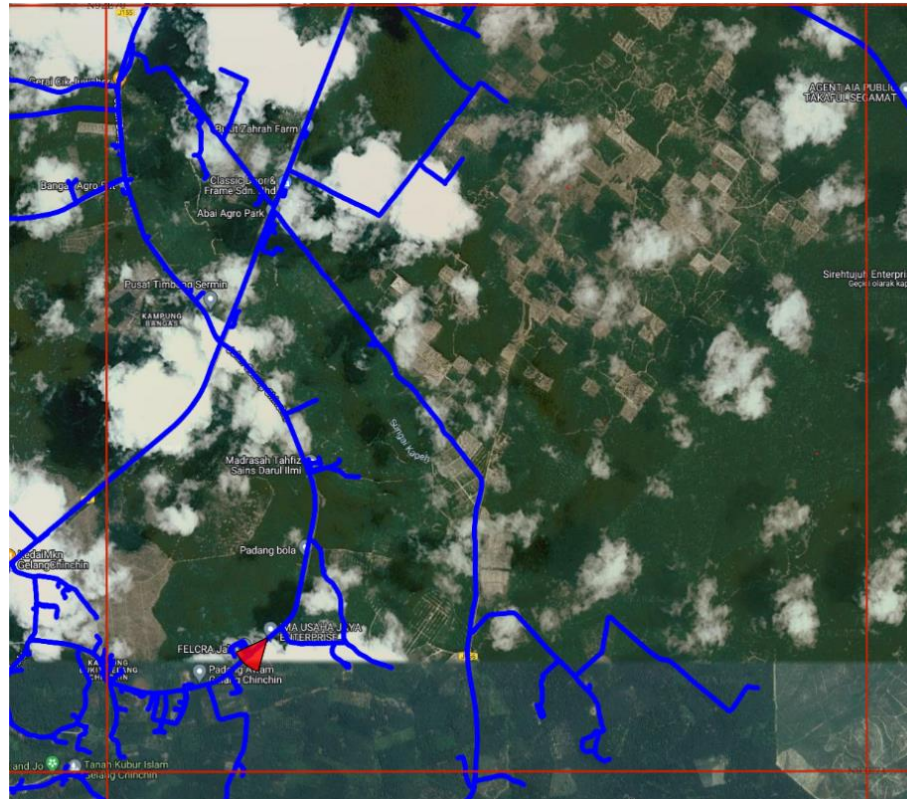
1. Open **RAIDRIVE**
2. Go to add → NAS → Select the option with cloud storage
3. For Drive → selection drive → Y: *Example*
4. Host address → <https://host.example.com>
5. Port :5050 *example*
6. User Account and password : *registered with the server*
7. Click to connect, then transfer the files along with metadata into the NAS folder
8. The data will be shown in the **USV Web player**.

USV PRO

Example of the MMS data that has been upload & publish to a web player: **USVPRO**

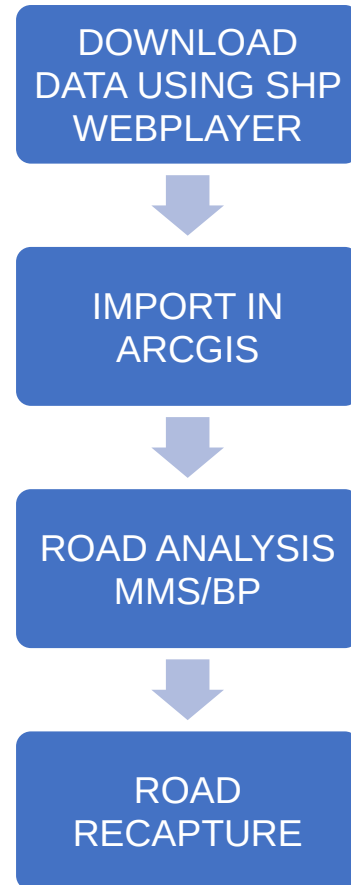
On the left side →
MMS/BP Tracklog

On the right side →
360 panoramic viewer



ROAD ANALYSIS

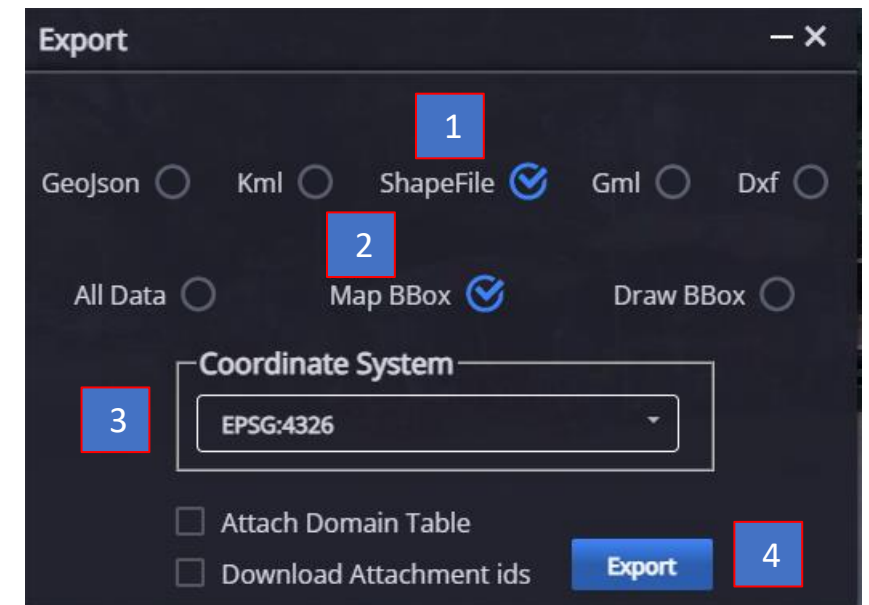
Road analysis diagram



EXPORTING LAYER – SHP FILE

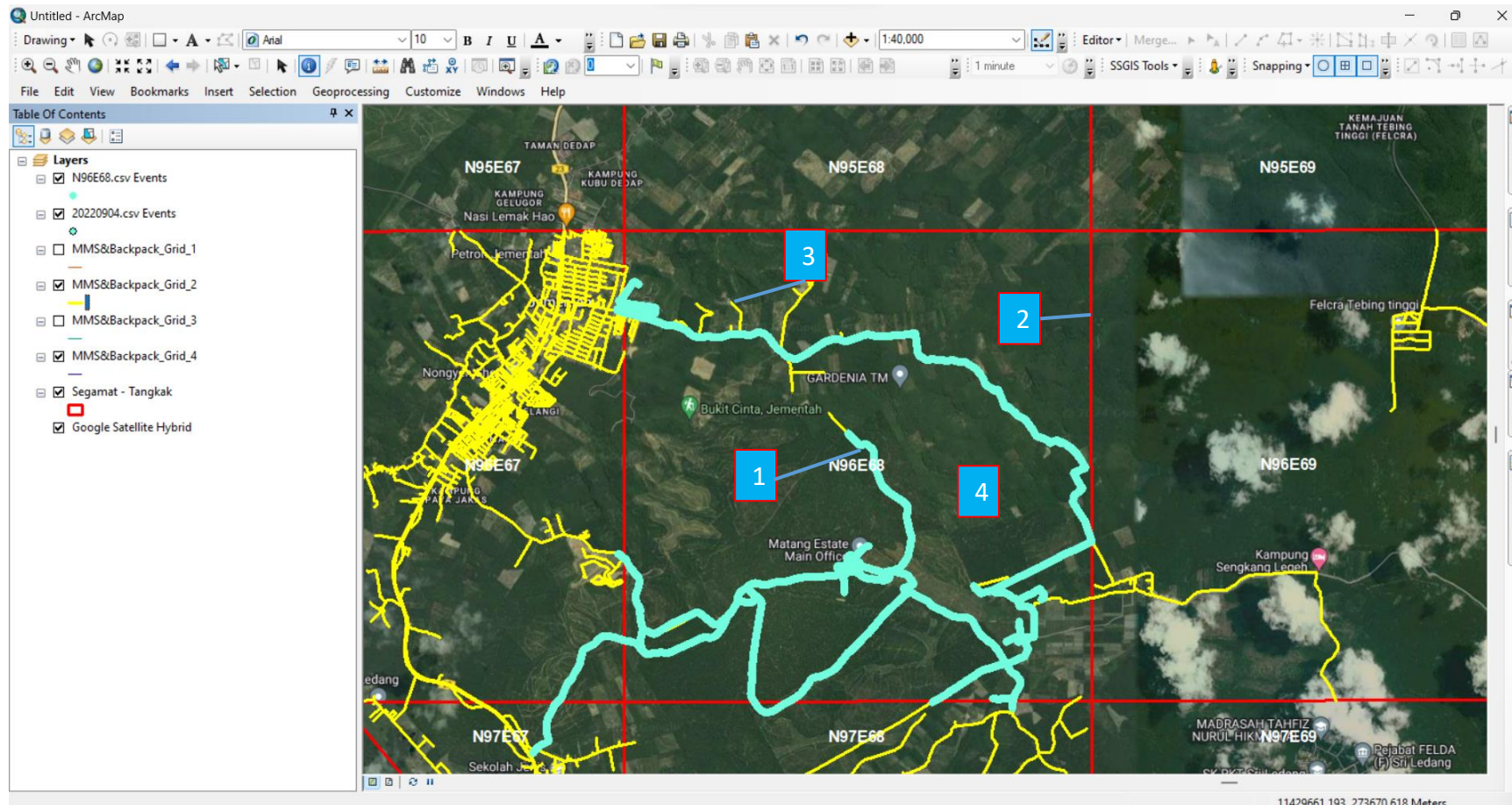


1. Select file format → *Shapefile*
2. Area of coverage to download :
All Data , Map BBox, Draw BBox
3. Coordinate system → *select*
EPSG:4326





ARCGIS – IMPORT SHP FILE



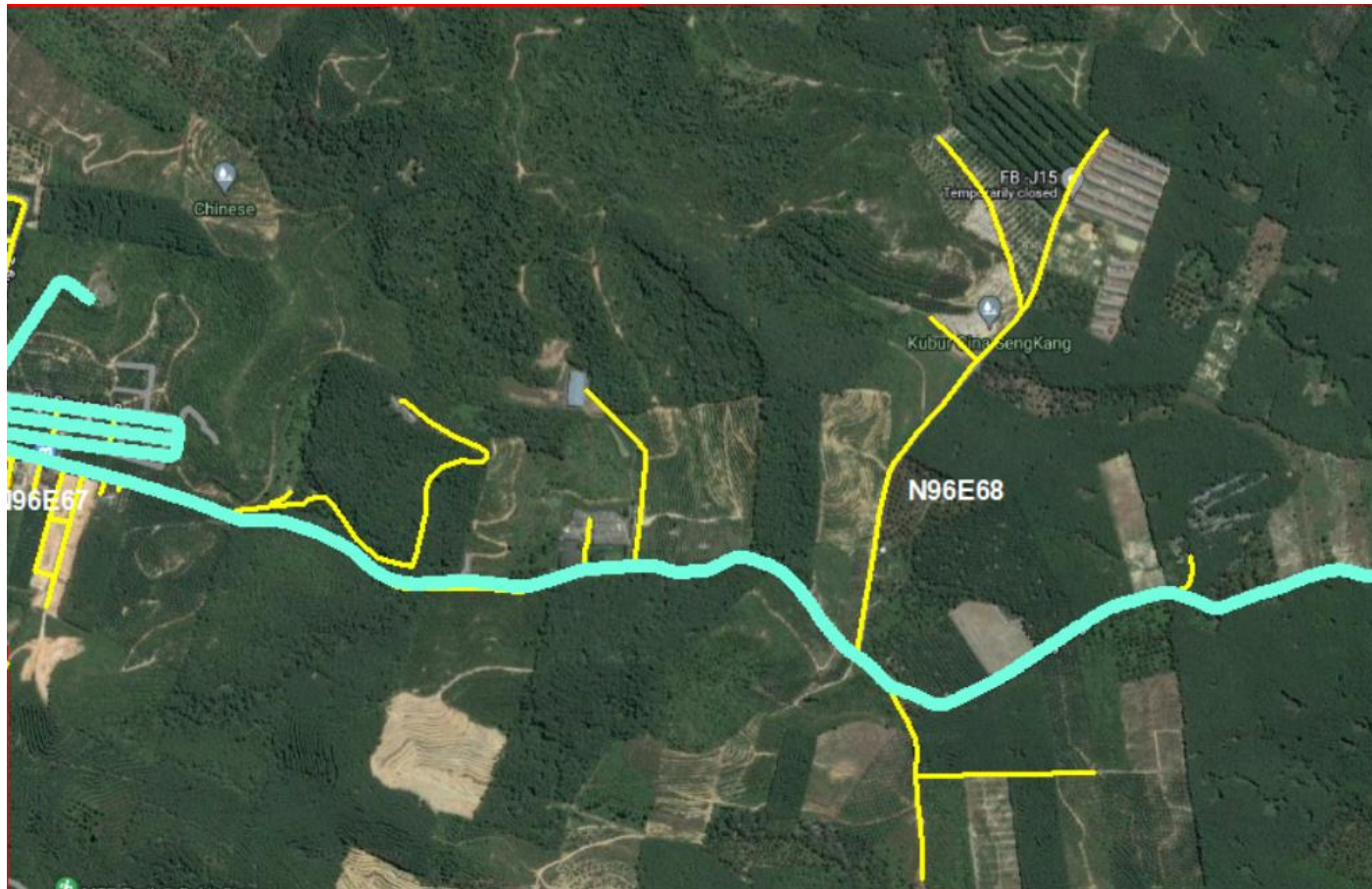
Data in Arcgis including:

1. Mms/Backpack
2. Grid 5km x 5km
3. Route Analysis
4. Base map

- MMS/BP data captured
- MMS/BP Analysis

For an example, data in Arcgis showing that in GRID_2 N96E68 still not complete and need to do a recapture. We will contact a site team to clarify the situation and the road condition. Somehow, it may not be accessible or any other issue related.

DATA RECAPTURE



Data recapture :

1. Road location that need to recapture.
2. Preparing shp file and put some remarks for site teams including XY coordinates for start and end of road.
3. Remarks from site team if the area is not accessible.
4. For an Example, on the left picture show